



Full Length Article

Two-phase flow characteristics of high-temperature CO₂ and water and dynamic wettability variations in nano-scale coal pores with different sizes and shapes



Shunqing Ma ^{a b c}, Baiquan Lin ^{a b c} , Jiajia Zhao ^{a b c}, Xiangliang Zhang ^{a b c} , Qian Liu ^d, Ting Liu ^{a b c}

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Highlights

- Establishment of nanoscale pores with different shapes and sizes in coal.
- Explored the laws and mechanisms of dynamic changes in wettability within nanopores.

- The wetting mode of droplets in nanopores of different sizes and shapes has been studied.

Abstract

Coal, a typical porous medium, contains numerous nano-scale pores with varying sizes and shapes within its matrix. Since wettability variations in these pores are crucial for gas–liquid flow, they considerably affect the efficiency of coalbed methane (CBM) extraction. In this study, the morphologies of coal pores subjected to high-temperature and high-pressure CO₂ treatment were observed by means of scanning electron microscopy (SEM) and atomic force microscopy (AFM). Based on these observations, molecular dynamics simulations were conducted to construct several typical water-bearing pore models. With these models, the characteristics of gas–liquid flow and wettability variations during CO₂ injection were explored. The results reveal that the wetting mode of droplets is closely related to the pore size, with droplets that exist in a Type I wetting mode boasting better wettability than those that exist in a Type II wetting mode. During the transition from water-wet to gas-wet states, maintaining the continuous movement of droplets proves to be more challenging than detaching them from their initial adsorbed configurations. CO₂ displaces water molecules in pores through two distinct mechanisms (competitive wetting and high-speed impact). In pores with weak water-wet properties, CO₂ advances in a crescent shape; otherwise, it progresses in a planar shape. Due to their different patterns of wettability variation, the displacement rates of water molecules in connected pores are remarkably higher than those in semi-closed pores. These findings, which elucidate the characteristics of gas–water two-phase flow in nano-scale pores and the transition from water-wet to gas-wet states, are beneficial for grasping the reasons for CBM production enhancement after CO₂ injection at the nanoscale.



Keywords

Different types of nanopores; Wettability change; Two-phase flow; Molecular dynamics; CO₂ capture; CBM

1. Introduction

Despite the abundance of coalbed methane (CBM) resources in deep coal seams in China, the efficient extraction of CBM in China is severely restricted by the prevalent geological conditions of coal reservoirs, including high in-situ stress, low permeability, and strong adsorption capacity. Permeability enhancement techniques such as hydraulic fracturing, hydraulic slotting, and hydraulic punching have been widely applied in coal mine production and have achieved fruitful results. These techniques are regarded as important techniques for coal reservoir stimulation. To achieve a more uniform fracture network, recent studies have integrated hydraulic slotting with hydraulic fracturing, where the slots guide the initial fracture propagation and induce microcracks that extend and connect with the slots [1,2]. However, the injection of large volumes of water-based fracturing fluids in hydraulic fracturing can lead to water sensitivity and water-lock effects in coal seams, which adversely affect efficient migration of CBM. Compared with conventional fracturing techniques, liquid nitrogen (LN₂) fracturing offers certain advantages. First, LN₂ reduces clay swelling and water-lock effects in the reservoir [3,4]. Second, its low-temperature effect alters the mechanical properties of coal rock, lowers the fracture initiation pressure, widens primary fractures, and induces the formation and propagation of secondary fractures, thus creating more channels for CBM migration [5]. Nevertheless, the feasibility of large-scale application of LN₂ fracturing remains controversial. Key challenges include modeling two-phase nitrogen flow, optimizing injection parameters, managing abrasives in fracturing fluids, and improving supporting equipment, monitoring, and evaluation systems [[6], [7], [8]]. Thermal injection, another effective method for CBM production enhancement besides fracturing, is particularly suitable for reservoirs with low water content and desorption difficulties. Common thermal injection methods include steam injection and microwave heating. Zhao et al. performed desorption-diffusion experiments on coal samples at multiple scales under various conditions. Their experimental results disclose that temperature elevation markedly enhances methane desorption and diffusion, and high temperatures can substantially raise the cumulative desorbed volume of methane [9]. Microwave heating not only promotes CBM desorption but also effectively removes moisture from coal seams. Owing to these advantages, microwave heating has been adopted for mitigating water-lock issues in gas reservoirs [10]. As the moisture content decreases, the permeability and gas production of coal seams grow notably [11]. In addition, the microwave-induced thermal effect can reduce the total content of oxygen-containing functional groups in coal, thereby weakening its maximum adsorption capacity for CH₄ [12]. Although microwave heating boasts great potential, technical challenges such as the delivery of microwaves into boreholes remain to be

addressed, and its field application and cost-effectiveness are yet to be fully validated. In contrast, injecting high-temperature CO₂ into high-gas and low-permeability coal seams (CO₂-ECBM) is a promising method coupling displacement, thermal, and dissolution effects. This technique effectively overcomes the limitations of strong adsorption and poor permeability and thereby considerably improves the efficiency of CBM extraction. Moreover, hot flue gas emitted from gas-fired power plants provides a stable and economical source of high-temperature CO₂. Its application not only improves economic benefits but also helps lower greenhouse gas emissions, which contributes to the mitigation of global climate change and the realization of China's "emission peak and carbon neutrality" goals.

Coal, a typical porous medium, contains numerous nano-scale pores with varying sizes and shapes. These pores, which possess a large specific surface area, provide sufficient storage space for gases. The wettability within these pores plays a crucial role in the adsorption, diffusion, and flow behaviors of gases [[13], [14], [15], [16], [17]]. In view of this fact, how CO₂ influences the wettability in coal pores under different CO₂ injection conditions has been extensively studied. In these studies, the contact angle measurement method is widely adopted due to its reliability, simplicity, and universality [18,19]. Compared to gaseous CO₂, supercritical CO₂ (sc-CO₂) demonstrates stronger wettability competition with water in coal pores, which can be attributed to the pressure- and temperature-dependent density of CO₂ [20]. Under typical reservoir conditions, the CO₂ wettability of coal increases with the rise of pressure and salinity, decreases with the rise of temperature, and has nothing to do with coal rank [21]. The CO₂ injectability varies in different reservoirs, and the transition from water-wet to gas-wet states requires overcoming a CO₂ pressure threshold [22]. In addition, some coal reservoirs are enriched with various organic minerals [23,24]. At low temperatures and high pressures, aside from competing with water for wetting, CO₂ also dissolves in water to form carbonic acid, which can dissolve minerals to some extent. This effect causes a decrease in hydrophilic components on pore surfaces and a deterioration in water wettability [25,26]. Low-field nuclear magnetic resonance relaxation time analysis technique enables the measurement of the relaxation times of hydrogen-bearing fluids (¹H) in micro-pores in coal samples, facilitating the study on wettability variations at a microscopic scale [[27], [28], [29], [30]]. Yao et al. examined the influence of water occurrence state on wettability during CO₂-ECBM, and found that the presence of non-adsorbed water in coal dramatically reduces the pressure sensitivity and absolute value of CO₂ wettability [31]. After identifying the forms of water occurrence by taking relaxation times as the criterion, Zhu et al. reached the conclusion that that CO₂ inhibits the transformation of free water into adsorbed and capillary water, but its effect is more substantial effect on adsorbed water [32].

With the continuous development of computational power, methods such as finite element analysis and molecular simulation have provided microscopic insights into the mechanisms of CO₂ injection-induced wettability variations within pores [33].

Through the combination of the contact angle measurement method with the volume-of-fluid (VOF) method, Peng et al. discovered that weak water-wet conditions correspond to the latest breakthrough time but the highest displacement efficiency [34]. Ma et al. constructed slit-shaped pores with rough surfaces by molecular simulation, and noted that raising the injection pressure of hot flue gas is beneficial for the transition from water wet to gas wet states and the elimination of the water-lock effect [35]. Xu et al. developed an oil-gas-water-wall four-phase model, and succeeded in visualizing the microscopic process of shale oil displacement by CO₂ and the migration and distribution characteristics of fluids in pores [36].

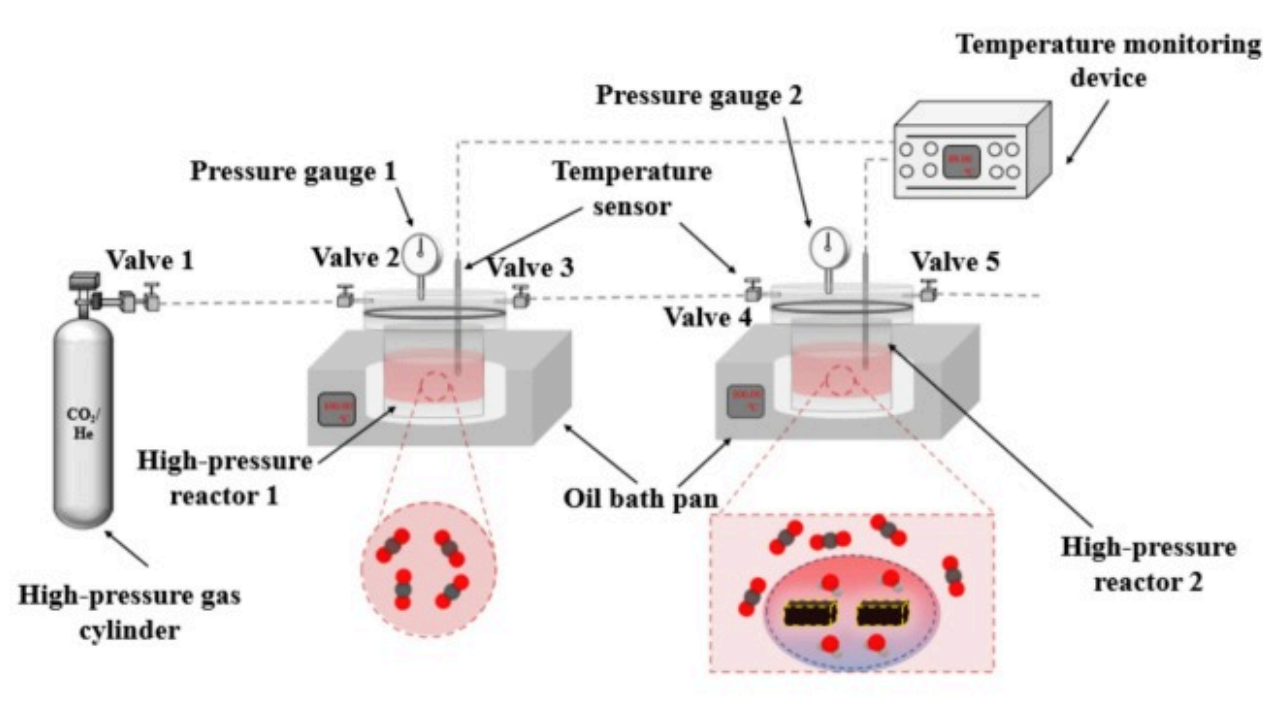
In summary, despite extensive experimental and simulation research on competitive wetting in the CO₂-ECBM process, the temperatures of CO₂ injected into coal seams in previous studies (below 323.00K) are much lower than those of high-temperature CO₂ in hot flue gas (approximately 448.00K). The mechanisms by which high-temperature CO₂ changes the wettability within nano-scale pores remain unclear. Moreover, pore shape and size have a considerable influence on wettability [[37], [38], [39], [40]]. To be specific, smaller pore diameters are associated with stronger capillary phenomena and more pronounced water-lock effects. However, most research to date has been dedicated to straight slit-shaped pores. In this study, the morphologies and size distributions of coal pores subjected to high-temperature and high-pressure CO₂ treatment were observed with the aid of scanning electron microscopy (SEM) and atomic force microscopy (AFM). Based on these observations, molecular dynamics simulations were conducted to construct several typical water-bearing pore models. With these models, the characteristics of gas-liquid flow and wettability variations during CO₂ injection were explored.

2. Experimental and simulation methods

2.1. Sample processing

After being collected from Wulanmulun (WLML) Mine in Inner Mongolia, China, bulk raw coal was crushed and further processed into cubic samples (5.00mm×3.00mm×2.00mm) using a wire cutting machine in the laboratory. Subsequently, argon ion polishing was performed on the upper surface of each cubic sample to achieve a comparable initial roughness. The high-temperature and high-temperature water-rock reaction system is depicted in Fig. 1. Prior to the experiment, the air tightness of the entire system was thoroughly tested. To ensure accurate and stable temperature and pressure of CO₂ during the experiment, Autoclave #1 was used to preheat and pressurize CO₂ to the target values (448.00K, 7MPa). The temperature sensor placed inside Autoclave #1 served to ensure that the gas temperature in the autoclave did not fall below the target temperature (as a result of limited heat transfer efficiency) when the methyl silicone oil in the oil bath had reached the setpoint. Autoclave #2 was

responsible for completing the water–rock reaction in a high-temperature and high-pressure CO₂ atmosphere. To maintain stable temperature and pressure during CO₂ transfer from Autoclave #1 to Autoclave #2, Autoclave #2 was also placed in the oil bath. Both autoclaves were equipped with pressure gauges to monitor the CO₂ pressure in real time. The argon ion-polished coal samples were placed at the bottom of Autoclave #2 and fully immersed in deionized water (150mL in this study). The reaction proceeded for 72h. After the reaction, the coal samples were dried in a drying oven at 45 °C for 24h. After cooling completely, they were stored for further use.



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Fig. 1. High-temperature and high-temperature water–rock reaction system.

2.2. SEM and AFM

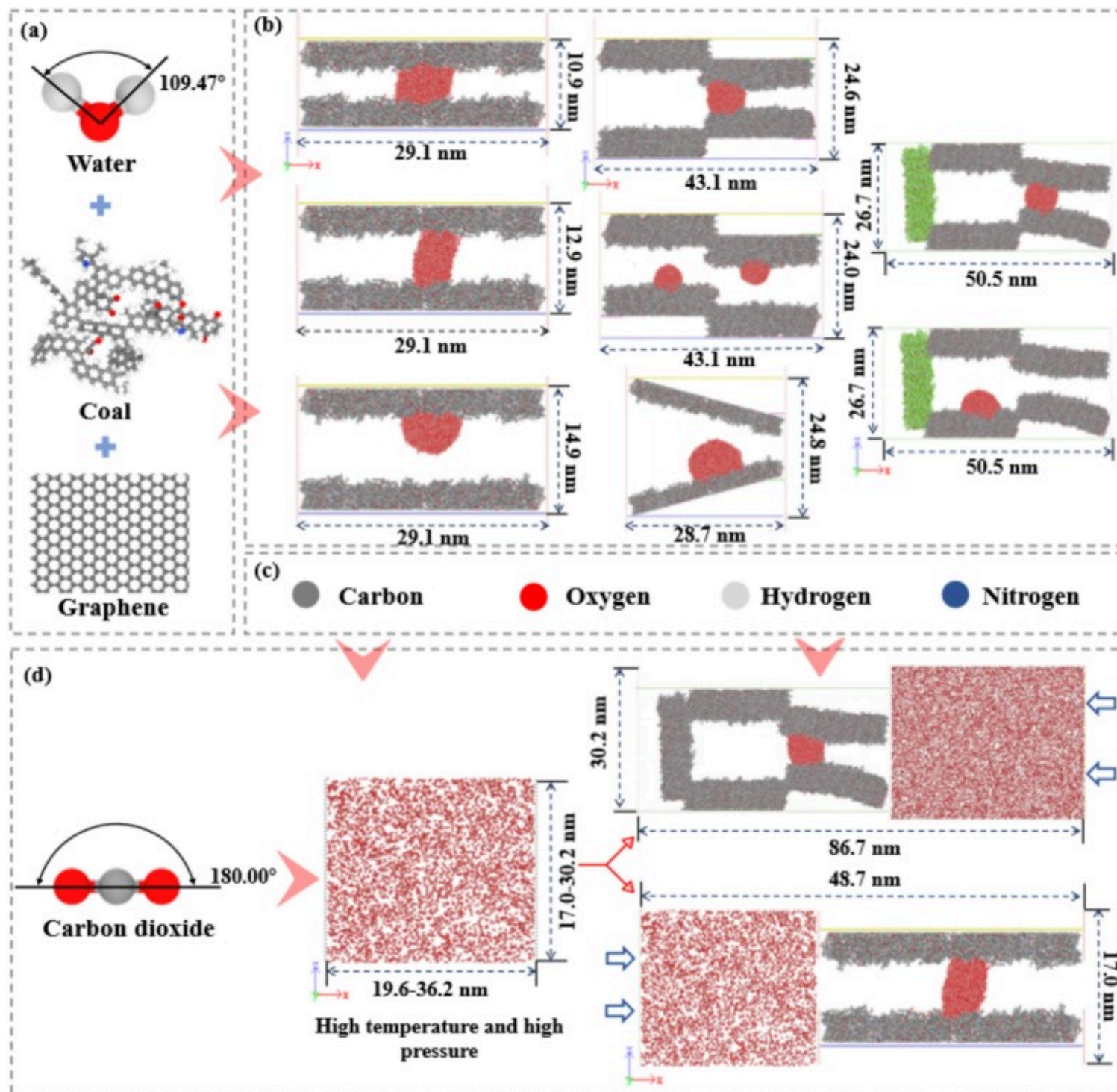
In this study, the pore-fracture network on the coal surface subjected to high-temperature and high-pressure CO₂ treatment was scanned with the aid of a BRUKER Dimension Icon atomic force microscope in the PeakForce Tapping mode, in the hope of

measuring the surface roughness. Each AFM scan was performed over a 10.00 $\mu\text{m} \times 10.00 \mu\text{m}$ area with a resolution of 512 sampling points per line. Meanwhile, the morphology and size distribution of nanoscale pores on the coal surface subjected to high-temperature and high-pressure CO₂ treatment were observed by a Verios 5UC scanning electron microscope.

2.3. Molecular dynamics simulations

2.3.1. Force field and model construction

As illustrated in Fig. 2, the DREIDING force field [41,42] was employed to describe anthracite. This force field, which can provide reliable estimates of coal substrate density and energy, has been commonly employed in related studies. The structure and composition of these macromolecules (molecular formula: C₁₉₉H₁₄₆N₂O₉) had been validated by ultimate analysis, Fourier transform infrared spectroscopy (FTIR), and Carbon-13 nuclear magnetic resonance spectroscopy (¹³C NMR) on actual samples. Pores with different shapes and sizes were constructed in the NPT ensemble using graphene plates as templates, with temperature regulated by a Nosé-Hoover thermostat and pressure controlled along the z-axis. To simulate real reservoir conditions, temperature and pressure were set to 333.00K and 25.00MPa, respectively. The constructed pores measured 28.70–50.50nm long, 10.90–26.70nm high, and 9.00nm wide. Water molecules were modeled using the SPC/E force field [43], with 5000 water molecules in three slit-shaped pores and 10,000 water molecules in others. Aiming at comprehensively reflecting the migration characteristics of droplets in semi-closed pores during CO₂ injection, water molecules in ink-bottle-shaped pores were placed in two modes: 1) Uniform placement in the central part along the Z-axis, and 2) Placement in the cavity at the rear. The EPM_2 force field considering the polarization effect was used to describe the movement of high-temperature and high-pressure CO₂ in pores. Different numbers of CO₂ molecules were placed in simulation boxes (19.60–36.20nm long, 17.00–32.20nm high, and 9.00nm wide) for the purpose of ensuring a consistent injection pressure (7MPa) in different pores and a sufficient gas source.



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Fig. 2. Model construction and operation.

2.3.2. Molecular dynamics simulations

The entire simulations were performed in the LAMMPS package [44]. A grouped relaxation approach was adopted: 1) High-temperature CO₂ was separately relaxed at 448.00K for 2.00ns in the NVT ensemble to obtain a reasonable configuration. 2) Water in pores was relaxed for 6.00ns in the NVT ensemble to allow it to wet the coal surface. Afterwards, the relaxed CO₂ and the relaxed water-containing pore model were combined and then subjected to the NVE ensemble, and a force was exerted on the graphene plate located either on the left or right to facilitate CO₂ injection. During the injection process, the Shake algorithm was employed to keep water molecules rigid [45]. The non-bonded interactions between atoms were described using the Lennard-Jones (12–6 LJ) potential and short-range Coulombic forces, both with a cutoff radius of 1.20nm [46]:

$$U(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q^j q^i}{4\pi\epsilon_0 r_{ij}} \quad (1)$$

where r is the distance between two atoms; ϵ is the depth of the potential well; and σ is the distance at which the LJ potential equals zero. The particle–particle–particle mesh (PPPM) solver was utilized to calculate long-range Coulomb forces (computational accuracy 1×10^{-4}) [47]. The Lorentz-Berthelot mixing rule was adopted to compute non-bonded interactions between different types of atoms:

$$\sigma_{\alpha\beta} = (\sigma_\alpha + \sigma_\beta)/2 \quad (2)$$

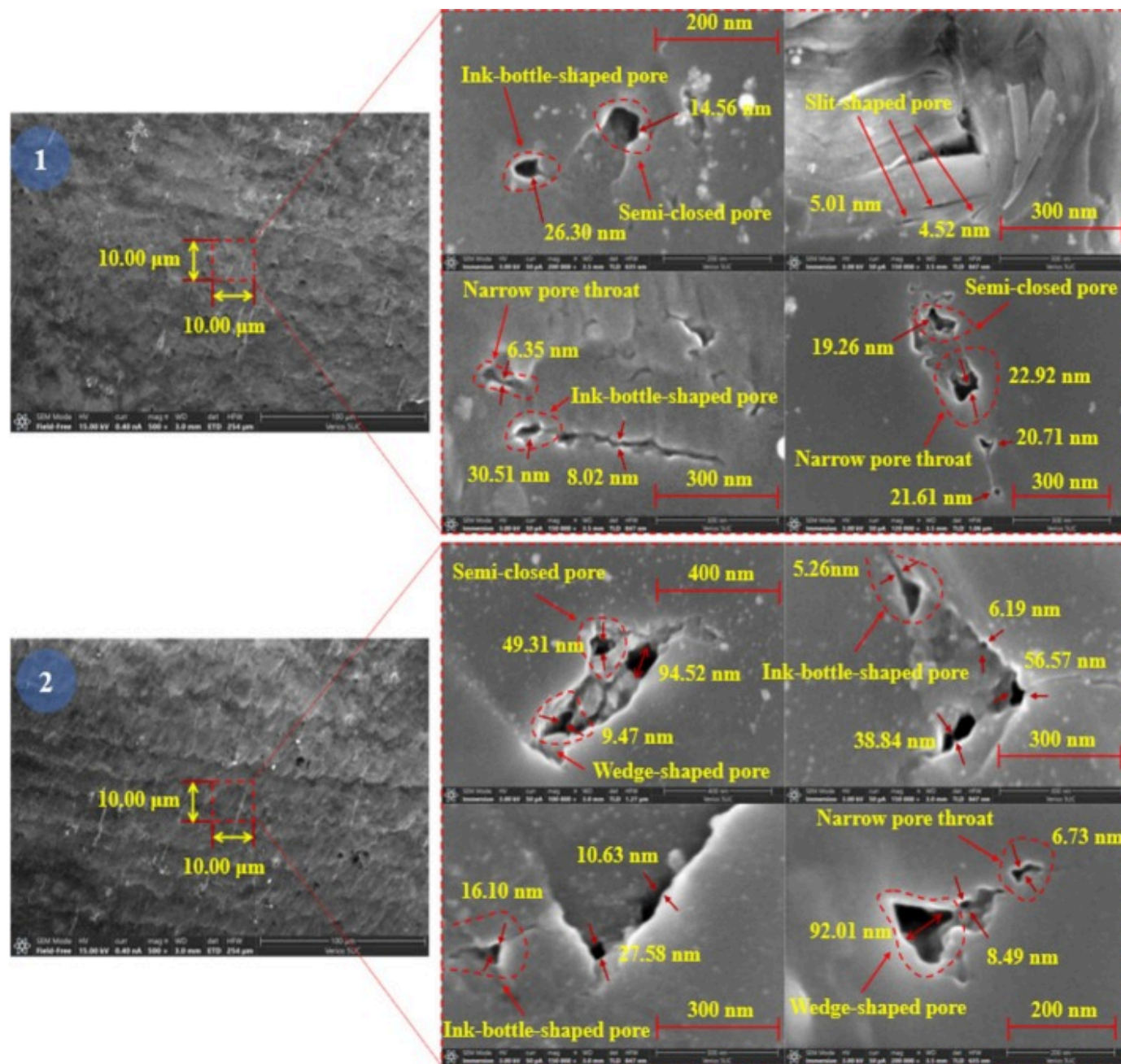
$$\epsilon_{\alpha\beta} = \sqrt{\epsilon_\alpha \epsilon_\beta} \quad (3)$$

where α and β are two different types of atoms. All model boundaries were periodic (PPP), and the coal body remained fixed throughout the simulation.

3. Results and discussion

3.1. Nanopore distribution on the coal surface after high-temperature CO₂ treatment

The nanopore distribution on the coal surface after high-temperature CO₂ treatment is presented in Fig. 3. At magnifications ranging from 1.00×10^5 to 2.00×10^5 , a variety of pores are clearly visible, including different-sized slit-shaped pores, different-shaped pores with narrow throats, and semi-closed pores predominantly featuring ink-bottle morphologies [[48], [49], [50]]. The pore size distribution in both interconnected and semi-closed pores is relatively broad, ranging from 4.52 nm to 92.01 nm. Among semi-closed pores, the cavities of ink-bottle-shaped pores are relatively large (up to 92.01 nm in diameter), while their throats are much narrower (only 5.26–8.49 nm). Among interconnected pores, slit-shaped pores are predominantly distributed within a narrow size range (4.52–5.01 nm). The openings of wedge-shaped pores and the throats of abruptly constricted/expanded pores are of relatively small diameters (6.35–22.92 nm). Besides, they are characterized by considerable pore narrowing, which hinders gas and water migration and further limits the efficiency of CBM production.



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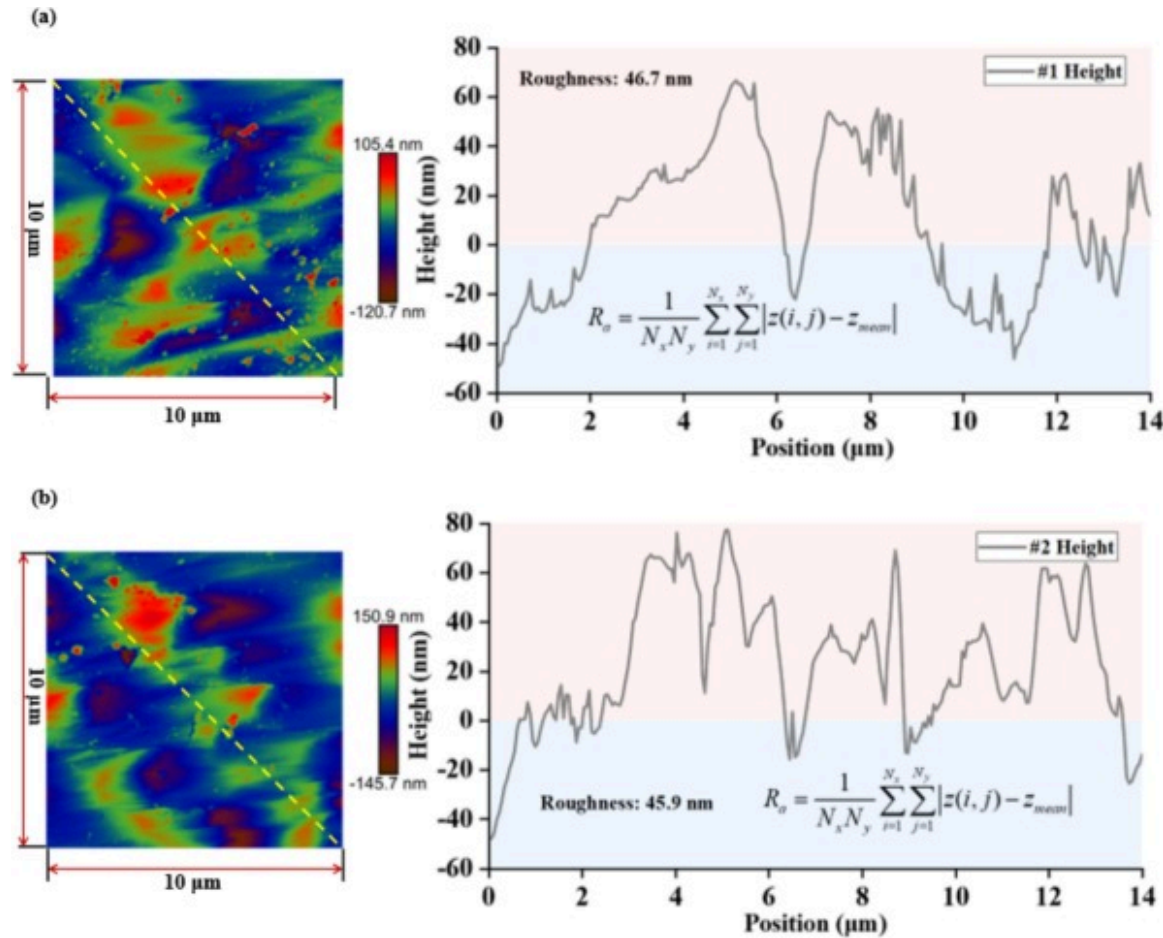
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Fig. 3. Typical nanopores.

Fig. 4 presents the AFM images of magnified areas from SEM images after high-temperature CO₂ treatment. Samples #1 and #2 display similar surface morphologies. After exposure to high-temperature CO₂, the coal surfaces are covered with a network of pores and fractures with diverse geometries and dimensions. The surface features well-developed peaks and valleys with distinct boundaries, broad aperture distributions, and intricate surface morphology. The average surface roughness (R_a) is calculated by Eq. (4):

$$R_a = \frac{1}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} |z(i, j) - z_{mean}| \quad (4)$$

where N_x and N_y are the number of scanning points along the X-axis and Y-axis, respectively; $z(i, j)$ is the height at Point (i, j) ; and z_{mean} is the average height of all measured points. The roughness values for Samples #1 and #2 are 46.7 nm and 45.9 nm, respectively.



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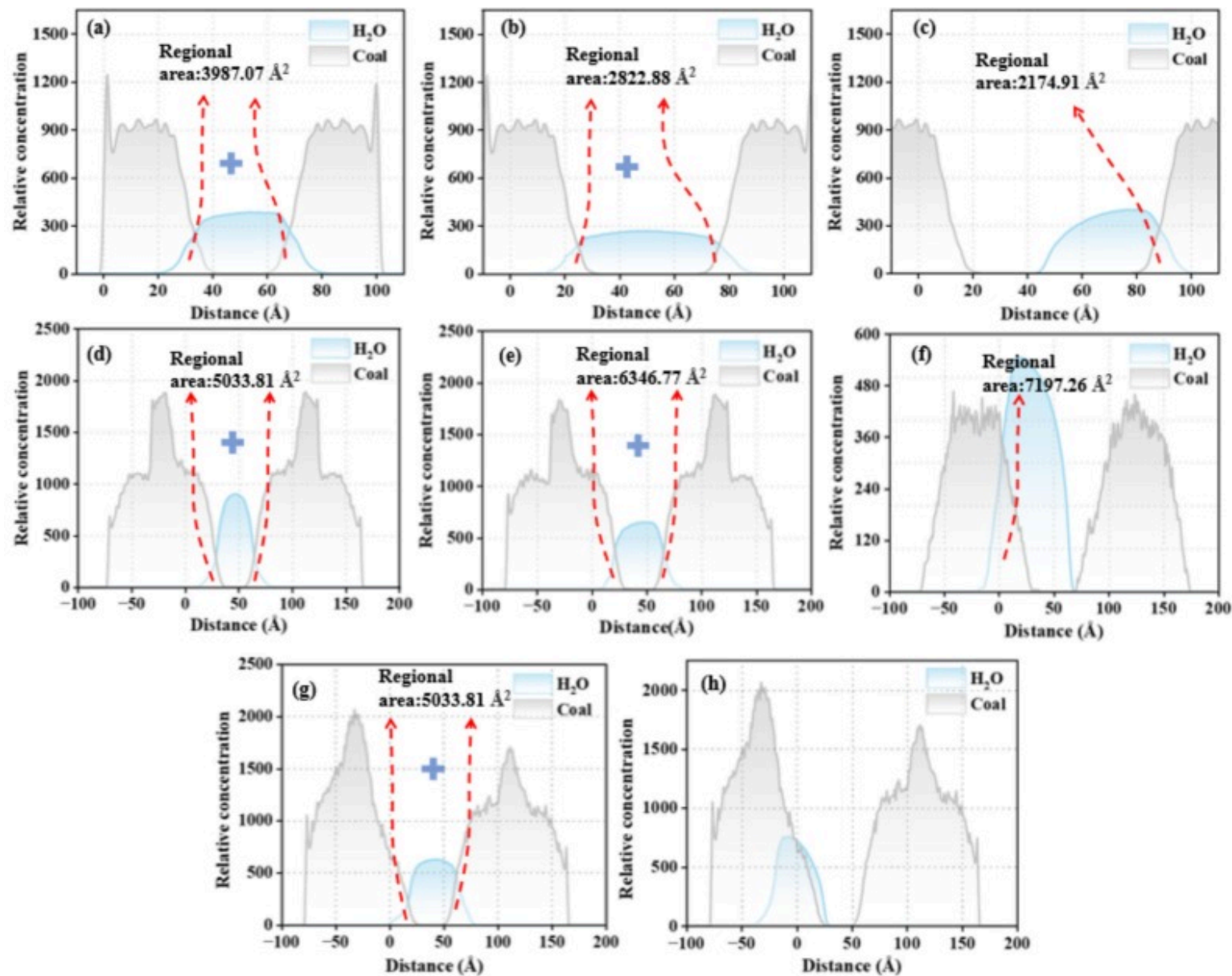
Fig. 4. Morphological features on the coal surface after high-temperature CO₂ treatment.

These pores, which possess a large specific surface area, provide sufficient storage space for gases. Changes in their internal wettability can notably influence the adsorption of gases, the mitigation of the water-lock effect, and the flow characteristics of the gas–water two-phase system. Therefore, in subsequent simulation work, nanopores with different sizes and shapes were established based on the SEM images in order to probe into the characteristics of gas–liquid flow and dynamic wettability variations during high-temperature CO₂ injection.

3.2. Wettability characteristics of droplets in pores with different sizes and shapes

3.2.1. Analysis on coal-water relative concentration distribution

It is challenging to accurately measure coal-water contact angles on rough coal surfaces. Compared to contact angle measurements, coal-water relative concentration distributions provide a more intuitive evaluation method [\[\[51\], \[52\], \[53\]\]](#). As given in [Fig. 5](#), the relative concentration distribution curves of coal and water overlap for pores with different shapes and sizes. It can be inferred that some water molecules penetrate the coal surface and enter the matrix, thereby facilitating water wettability. Generally, when the total number of water molecules is fixed, a larger overlapping area of the relative concentration distribution curves corresponds to better water wettability on the coal surface.



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Fig. 5. Coal-water relative concentration distributions in pores with different sizes and shapes.

As can be observed from Fig. 5(a) and (b), droplets in small- and medium-sized slit-shaped pores (4.00–6.00 nm) are compressed by the pore walls and can thus simultaneously wet both upper and lower surfaces (Type I wetting). However, as the pore size grows, droplets are continuously prolonged, and some water molecules that have penetrated the coal matrix are forced to migrate towards the coal surface under the action of hydrogen bonds between water molecules. As a result, the overlapping area of the relative concentration distribution curves shrinks from 3,987.07 Å² to 2,822.88 Å², a mark of poorer water wettability. When the pore size further grows to 8 nm, the distance between the lower coal surface and the droplet is too long, even exceeding the cutoff radius of van der Waals interactions. In this case, water molecules fail to simultaneously wet both surfaces, so that they adsorb on the upper surface in a quasi-spherical shape under the action of surface tension (Type II wetting). Under this configuration, the overlapping area of the relative concentration distribution curves is only 2,174.91 Å², representing the poorest wettability.

Water molecules exhibit distinct wetting behaviors in pores with relatively complex surfaces. As shown in Fig. 5(d)–(f), in the Z-shaped pore, water molecules aggregate into two droplets, each adsorbing on the nearest coal surface in a Type II wetting mode. Meanwhile, the overlapping area is relatively small (5,033.81 Å²). In the abruptly constricted/expanded pore, water molecules preferentially adsorb at the pore throat in a Type I wetting mode rather than in the larger pore cavity. Consequently, this type of pore displays a larger overlapping area and better wettability relative to the Z-shaped one. In the wedge-shaped pore, water molecules aggregate into a single large droplet that adsorbs on the coal surface in a Type II wetting mode, yet this type of pore demonstrates the largest overlapping area (7,197.26 Å²), which is attributed to the inclined angle of the coal surface. Such a result means that the overlapping area in the wedge-shaped pore alone cannot reliably assess the wettability of water molecules on the coal surface. The wetting behaviors of water molecules in semi-closed pores are provided in Fig. 5(g) and (h). In the Type I ink-bottle-shaped pore, water molecules adsorb at the bottleneck via Type I wetting. Compared to the throat of the abruptly constricted/expanded pore, the narrower neck of the ink-bottle-shaped pore compresses the droplet more intensely, which brings about a relatively large overlapping area (7,251.43 Å²). In the Type II ink-bottle-shaped pore, however, the relative concentration distribution no longer holds reference value due to the spatial coincidence between water molecules and the non-contact coal body.

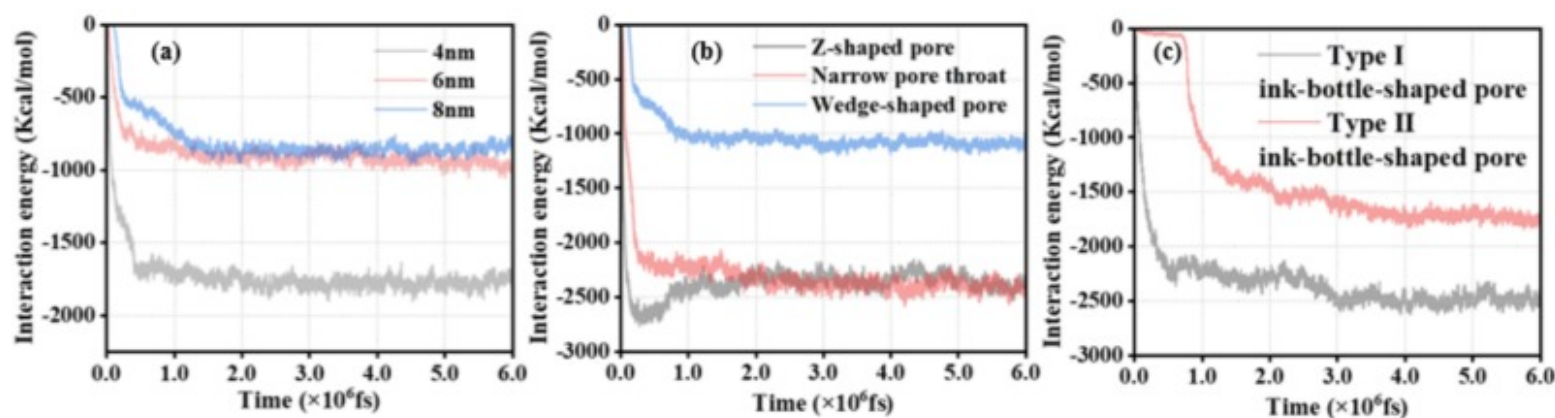
3.2.2. Coal-water interaction energy

To more comprehensively disclose the differences in wettability of water molecules in pores with different sizes and shapes and accurately analyze the changes in wettability when the coal-water relative concentration distribution becomes invalid, the coal-

water interaction energy was calculated by Eq. (5) for eight types of pores [54]:

$$E_{\text{interaction}} = E_{AB} - E_A - E_B \quad (5)$$

where E_{AB} is the total energy of the A-B complex; and E_A and E_B are the energies of substances A and B when they exist alone. A lower coal-water interaction energy is indicative of stronger interactions between them and better wettability of water on the coal surface. It can be seen from Fig. 6(a), as the size of slit-shaped pores expands, the interaction strength plunges, and the wettability within these pores deteriorates. This finding agrees with the conclusion drawn from the analysis on coal-water relative concentration distribution. Besides, droplets adsorbed in a Type I wetting mode are more sensitive to pore size variations.



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Fig. 6. Coal-water interaction energy in pores with different sizes and shapes.

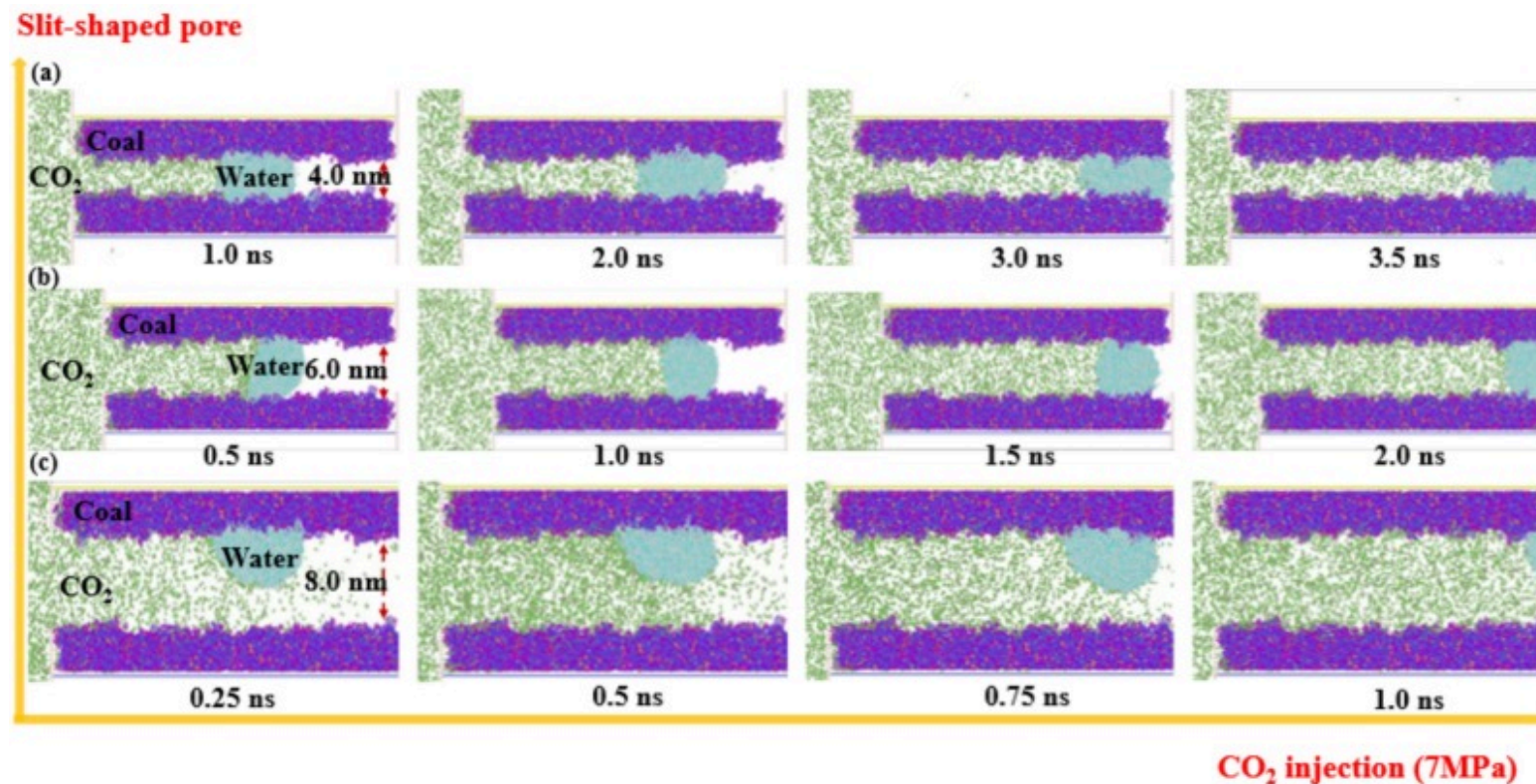
In pores with relatively complex surfaces (Fig. 6(b)), the coal-water interaction intensity in the abruptly constricted/expanded pore is slightly higher than that in the Z-shaped pore, though the difference is minimal. Meanwhile, both are substantially higher than that in the wedge-shaped pore. This is because the larger volume of the droplet in the wedge-shaped pore creates a greater density difference between the interior and surface of the droplet. Accordingly, outer water molecules tend to contract inward under the action of surface tension, which is inconducive to the full spreading of the droplet on the coal surface. In the Type I ink-bottle-shaped pore, water molecules adsorb at the bottleneck in a Type I wetting mode, demonstrating the strongest coal-water

interaction in this study. The coal-water interaction intensity in the Type II ink-bottle-shaped pore lies between those of the wedge-shaped pore and the Z-shaped pore.

3.3. CO₂-water two-phase flow characteristics and dynamic wettability variations in different-sized slit-shaped pores

3.3.1. Migration characteristics of droplets in different-sized slit-shaped pores

Under the impact of high-temperature and high-pressure CO₂, droplets within different-sized slit-shaped pores gradually leave the pore space, accompanied by a transition from water-wet to gas-wet states. The wetting behaviors of droplets within different-sized pores have an important effect on their migration modes. As depicted in Fig. 7(a) and (b), the droplets existing in a Type I wetting mode completely block the migration channel of CO₂ molecules, which creates a huge pressure difference on their two sides. This pressure difference disrupts the original adsorption configurations of the droplets and drives their collective migration toward the outlet of the slit-shaped pores. Moreover, the smaller the pore size, the stronger the resistance to droplet displacement, and hence the later the time when notable displacement occurs, as evidenced by the fact that the droplet in the pore with a diameter of 4.00nm begins moving 0.50ns later than the one in the pore with a diameter of 6.00nm. When the droplet exists in a Type II wetting mode (Fig. 7(c)), upon CO₂ impact, water molecules on the lower middle side of the droplet, which are more distant from the coal surface, shift to the right first. As a consequence, the droplet tilts to the right as a whole, forcing water molecules within the coal-water adsorption layer to detach from the coal body and return to the bulk phase. During the movement of the droplet, distinct advancing and receding contact angles are observable. Moreover, owing to the poor water wettability, the water molecules existing in a Type II wetting mode has almost completely exited the pore space by 1.00ns.



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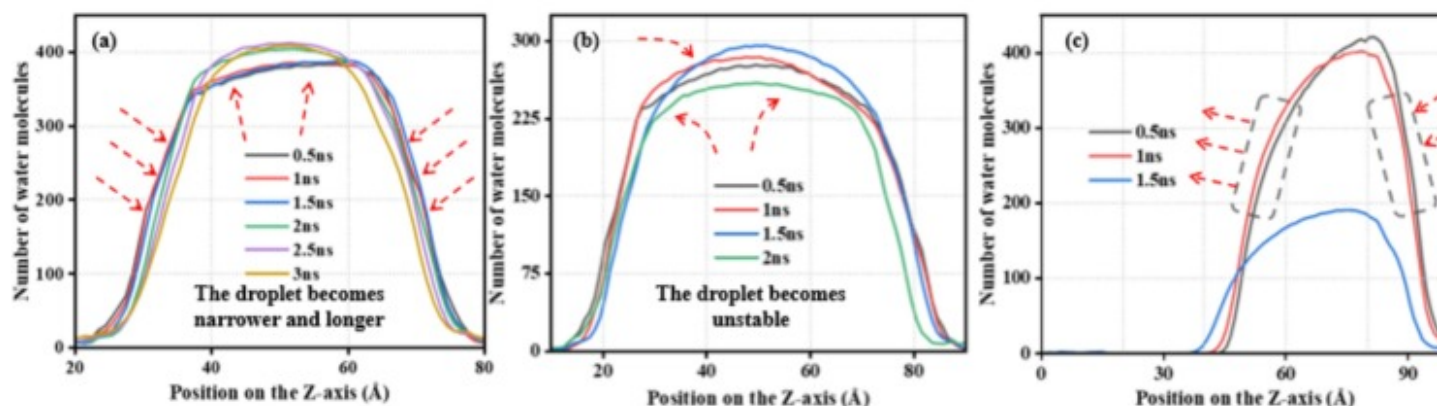
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Fig. 7. Movement of droplets in different-sized slit-shaped pores: (a) 4nm; (b) 6nm; (c) 8nm.

3.3.2. Morphological changes of droplets in different-sized slit-shaped pores

The migration of droplets in different-sized slit-shaped pores is accompanied by morphological changes which in turn affect the transition from water-wet to gas-wet states within these pores. How water molecules are distributed along the Z-direction at representative moments in different-sized slit-shaped pores is displayed in Fig. 8 [55,56]. In the small-sized pore (4.00nm), the coal-water interaction is relatively strong. The injection of high-temperature CO₂ into this pore reduces the number of water molecules in the water adsorption layers on the upper and lower pore walls and promotes them to move towards the middle of the droplet. This process elongates the droplet morphology and makes it more displaceable. In the medium-sized pore (6.00nm),

the droplet, particularly its central part that accommodates massive water molecules, keeps shaking and becomes unstable under the impact of high-temperature CO₂, which hinders the formation of a stable adsorption configuration at the coal-water contact surface and leads to a decrease in water wettability. In the large-sized pore (8.00nm), the injection of high-temperature CO₂ affects the morphology of the droplet the most significantly. The intense sweeping action causes the whole droplet to shift and interrupts hydrogen bonds between water molecules, so that many water molecules detach from the droplet and leaves the pore space. As shown in Fig. 8(c), the droplet volume at 1.50ns is much smaller than that at 1.00ns.



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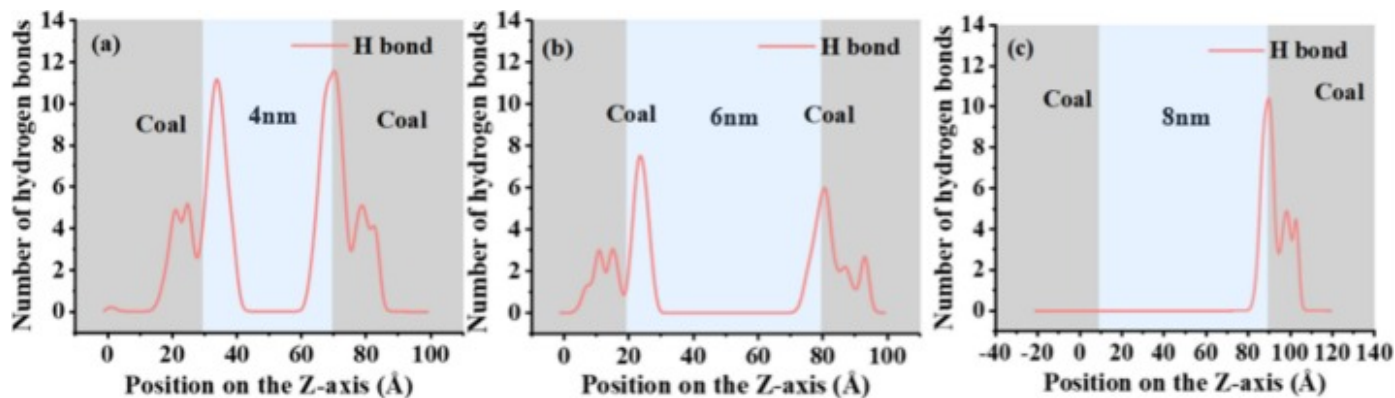
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Fig. 8. Spatial distribution of water molecules in different-sized slit-shaped pores.

3.3.3. Mechanism and dominant factors of CO₂ displacement of nanodroplets in slit-shaped pores

In different-sized slit-shaped pores, CO₂ molecules must disrupt the stable adsorption configuration of nanodroplets to achieve displacement. Whether this stable configuration can remain intact depends largely on coal-water hydrogen bonds and water-water hydrogen bonds. The formation of hydrogen bonds is conditional on specific distances and angles. The configuration of nanodroplets within slit-shaped pores play a decisive role in determining the formation of these bonds, and this configuration is in turn controlled by the width of slit-shaped pores. As the pore size grows, the droplet shape evolves from a flattened shape to an elongated one, accompanied by a transition from Type I to Type II wetting. The distribution of coal-water hydrogen bonds in different-sized slit-shaped pores is given in Fig. 9. When the pore width increases from 4.00nm to 6.00nm, hydrogen bonds maintain two peaks near the upper and lower pore walls, respectively, but the peak intensities fall obviously. As the pore size

increases to 8.00nm, water molecules can no longer simultaneously wet both walls. Instead, they are adsorbed near the upper wall in a quasi-spherical shape under the influence of surface tension, resulting in an evident hydrogen bond peak only near the upper wall.



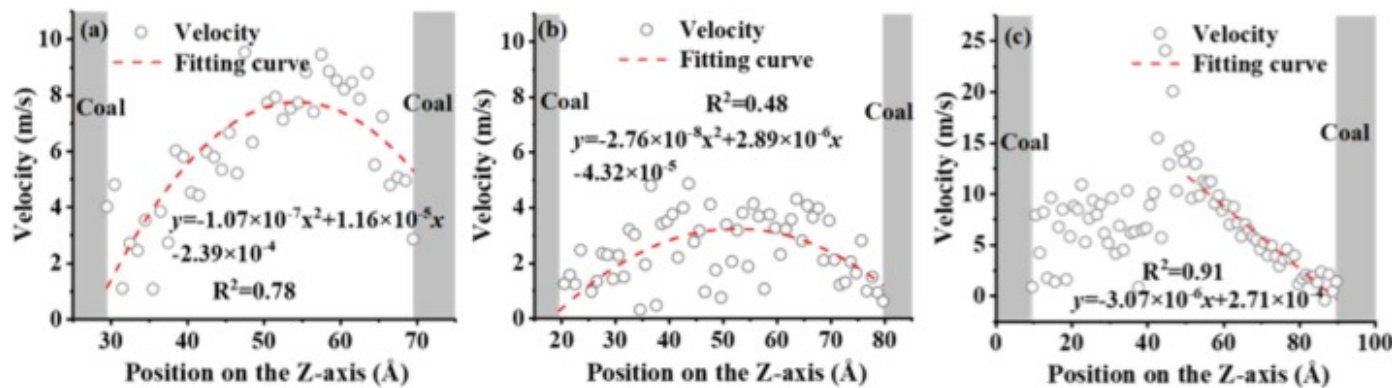
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Fig. 9. Hydrogen bond distribution in different-sized slit-shaped pores.

When high-temperature and high-pressure CO₂ molecules enter slit-shaped pores, they disrupt the stable configuration of nanodroplets mainly through two mechanisms: high-speed impact and competitive wetting. These two mechanisms act at different positions within the droplet. In general, when CO₂ molecules are located in the central pore space, they move rapidly, and water molecules remain stably confined due to intermolecular hydrogen bonding. In this case, high-speed impact dominates the displacement process. In contrast, when CO₂ molecules approach adsorption sites near the pore walls, their velocity decreases, and the stability of water molecules is maintained primarily by hydrogen bonds formed between electronegative atoms (e.g., nitrogen and oxygen) in coal and hydrogen atoms in water. In this case, competitive wetting is the dominant mechanism. Although competitive wetting displaces water molecules more effectively than high-speed impact, the displacement of adsorbed water molecules on the coal surface occurs more slowly than that in the bulk phase at the droplet center due to the stronger attractive forces near the pore wall. The velocity distribution of nanodroplets in different-sized slit-shaped pores are illustrated in Fig. 10. In the Type I wetting mode, the velocity of water molecules within the droplet varies quadratically with the increase of height along the Z-axis. Water molecules in the central pore space migrate faster, while those near the pore walls move more slowly. As the pore size increases, the displacement effects of high-speed impact and competitive wetting become

more consistent. In the Type II wetting mode, the velocity of water molecules within the droplet decreases linearly with the increase of height along the Z-axis (towards the upper pore wall). It is noteworthy that the velocity distribution within the 10–40 Å range is relatively disordered, which corresponds to free water molecules that have detached from the nanodroplet. Regardless of the wetting mode, as the water molecular layers in the nanodroplet move from the vicinity of the pore wall toward the pore interior, the dominant displacement mechanism transitions from competitive wetting to high-speed impact.



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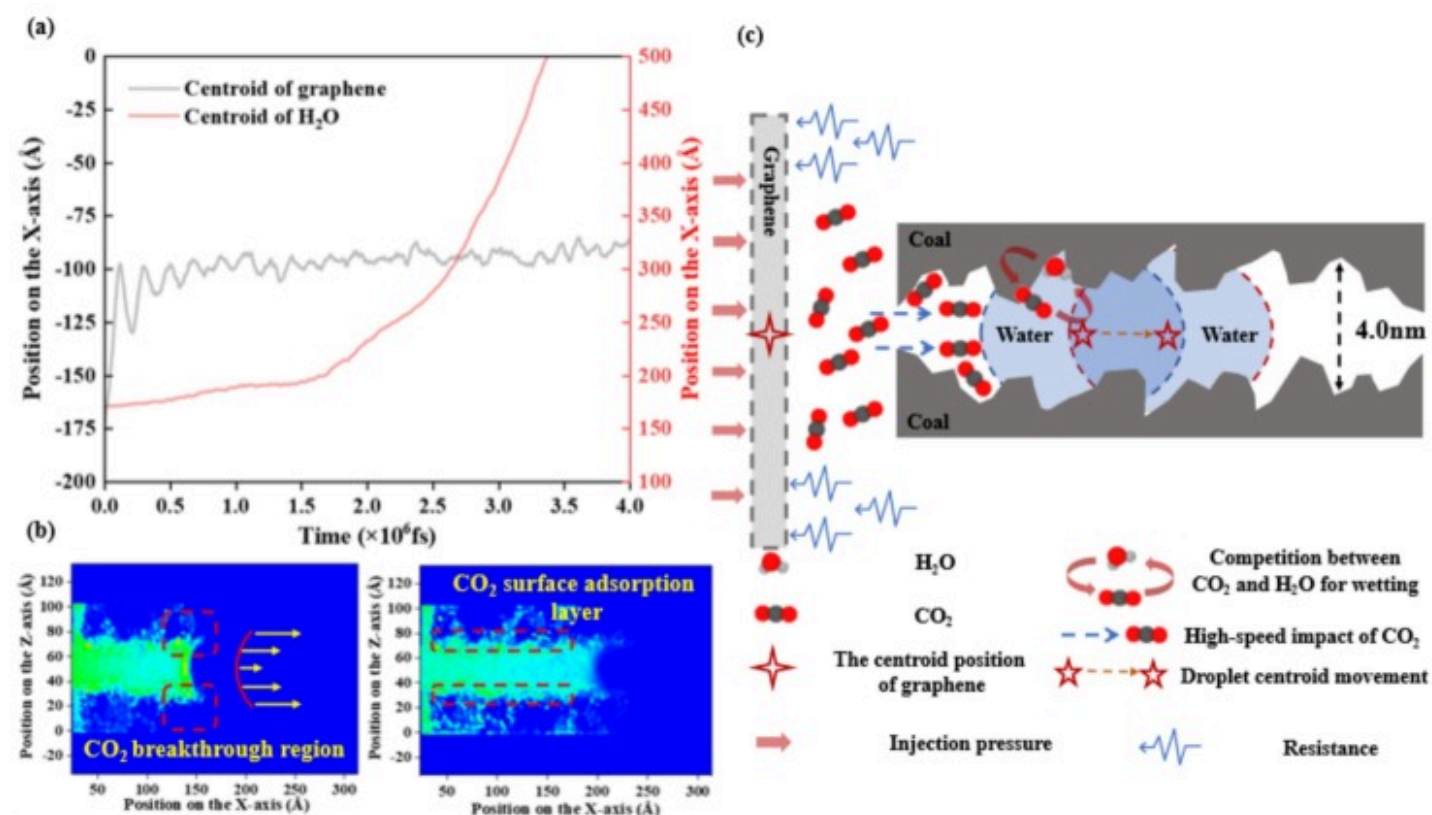
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Fig. 10. Velocity distribution of nanodroplets in different-sized slit-shaped pores.

3.3.4. Wettability variation and droplet resistance in different-sized slit-shaped pores

To more intuitively display the flow characteristics of the gas–water two-phase system within slit-shaped pores and evaluate the ease of transition from water-wet to gas-wet states, the study examined how piston wall oscillation, droplet centroid coordinates in the X-direction, and coal–water interaction energy vary over injection time, and analyzed the invasion characteristics of CO₂ in different-sized slit-shaped pores based on density contour maps. As illustrated in Fig. 11, high-temperature and high-pressure CO₂ shows poor injectability in the small-sized pore (4.00nm), as evidenced by violent piston wall oscillations in the X-direction in the initial period. After part of CO₂ molecules enter the pore, they adsorb on the upper and lower pore walls to form high-density adsorption layers. In this case, CO₂ molecules advance into the pore in a crescent shape. As the pore size increases (6.00nm, Fig. 12), the injectability of high-temperature and high-pressure CO₂ improves in the initial period, but the piston wall still oscillates slightly in the X-direction because the droplet still completely blocks the molecular transport channel. As the pore

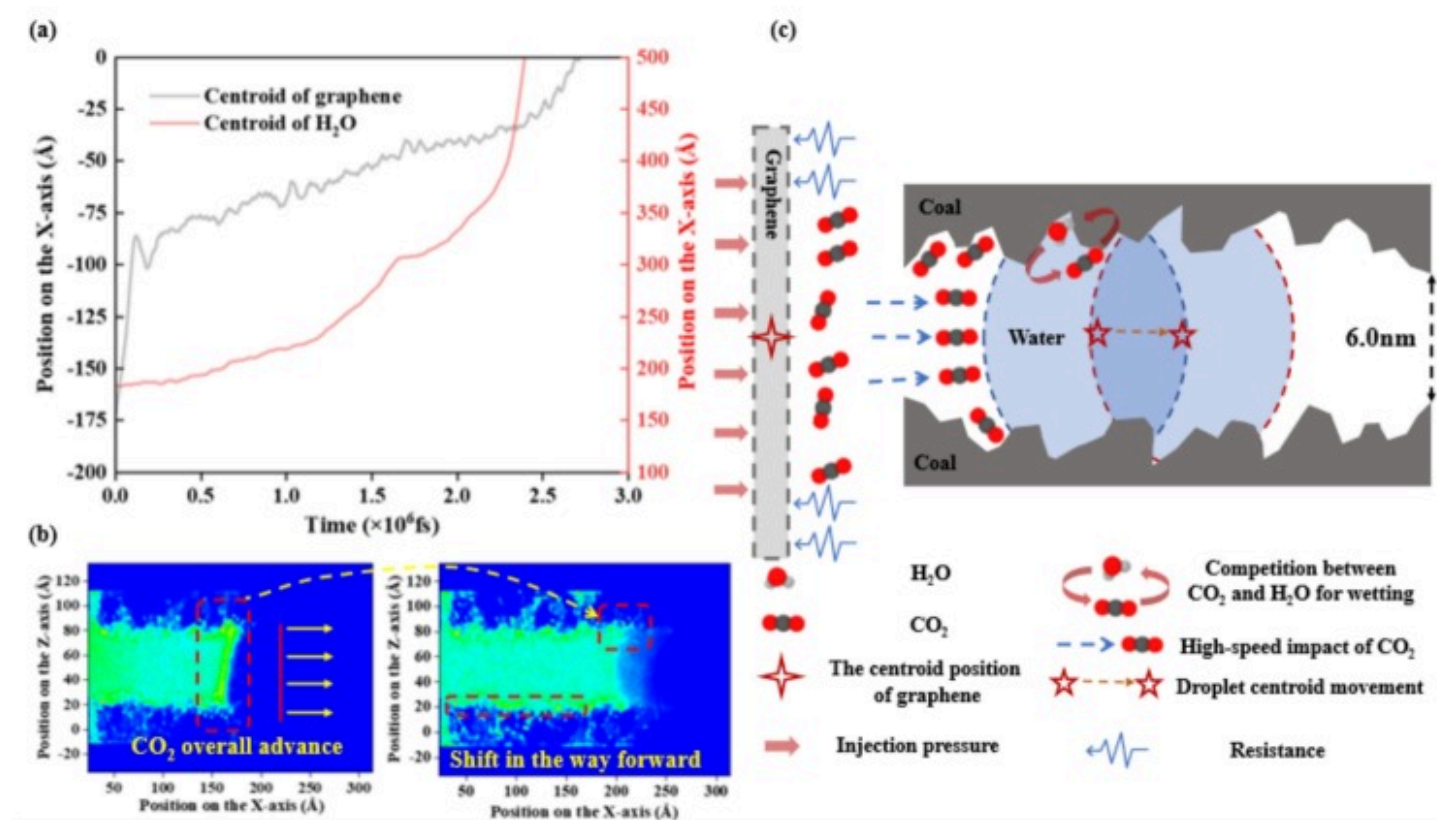
size expansion deteriorates the initial water wettability, the effectiveness difference between two displacement mechanisms, i.e., rapid impact in the central part versus competitive wetting on the surface, becomes negligible, and CO₂ advances into the pore in a planar manner. It is noteworthy that as the droplet is displaced from its original adsorption site, water molecules continuously contact and adsorb on the coal surface ahead. Consequently, it gets harder to continuously push the droplet, and CO₂ starts to advance in a crescent shape. In the large-sized pore (8.00 nm, Fig. 13), the presence of a larger pore volume and connected flow channel dramatically promotes the injectability of CO₂ molecules in the initial period, and eliminates piston wall oscillations in the X-direction. Compared to droplets in small- and medium-sized pores, the droplet in the large-sized pore exhibits earlier and faster centroid displacement in the X-direction. While moving in small- and medium-sized pores, the droplet maintains a nearly constant volume and exerts some resistance to invading CO₂ molecules. In contrast, it loses numerous water molecules when moving towards the opening of the large-sized pore. These rapidly escaping water molecules induce the high-speed rightward displacement of the droplet centroid.



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Fig. 11. Injection characteristics of CO₂ and its competitive wetting with the nanodroplet in the 4nm slit-shaped pore: (a) Centroid variations of graphene and nanodroplet during CO₂ injection; (b) Density distribution of CO₂ within the 4nm slit-shaped pore during CO₂ injection; (c) Gas-water flow characteristics and wettability change mechanism in the 4nm slit-shaped pore.

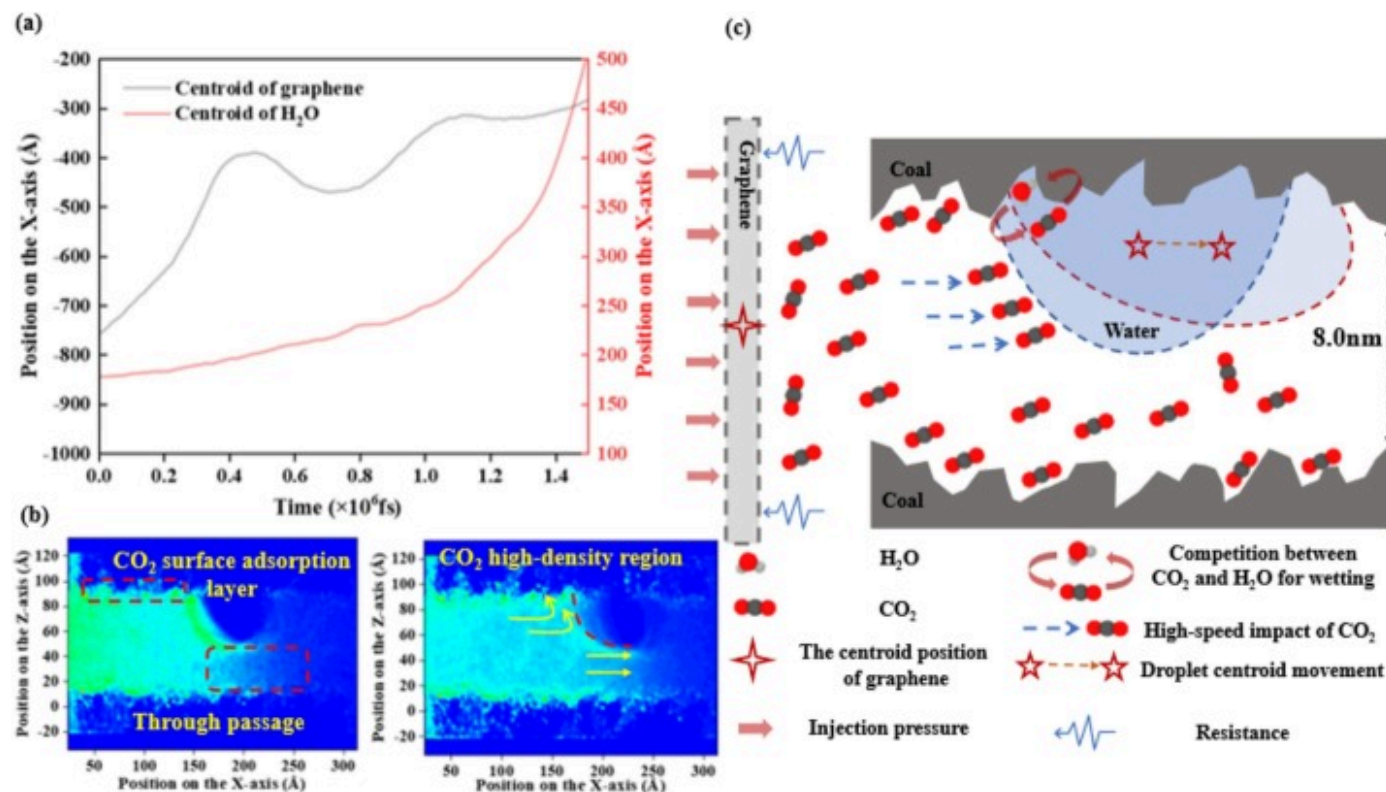


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Fig. 12. Injection characteristics of CO₂ and its competitive wetting with the nanodroplet in the 6nm slit-shaped pore: (a) Centroid variations of graphene and nanodroplet during CO₂ injection; (b) Density distribution of CO₂ within the 6nm slit-

shaped pore during CO₂ injection; (c) Gas-water flow characteristics and wettability change mechanism in the 6nm slit-shaped pore.



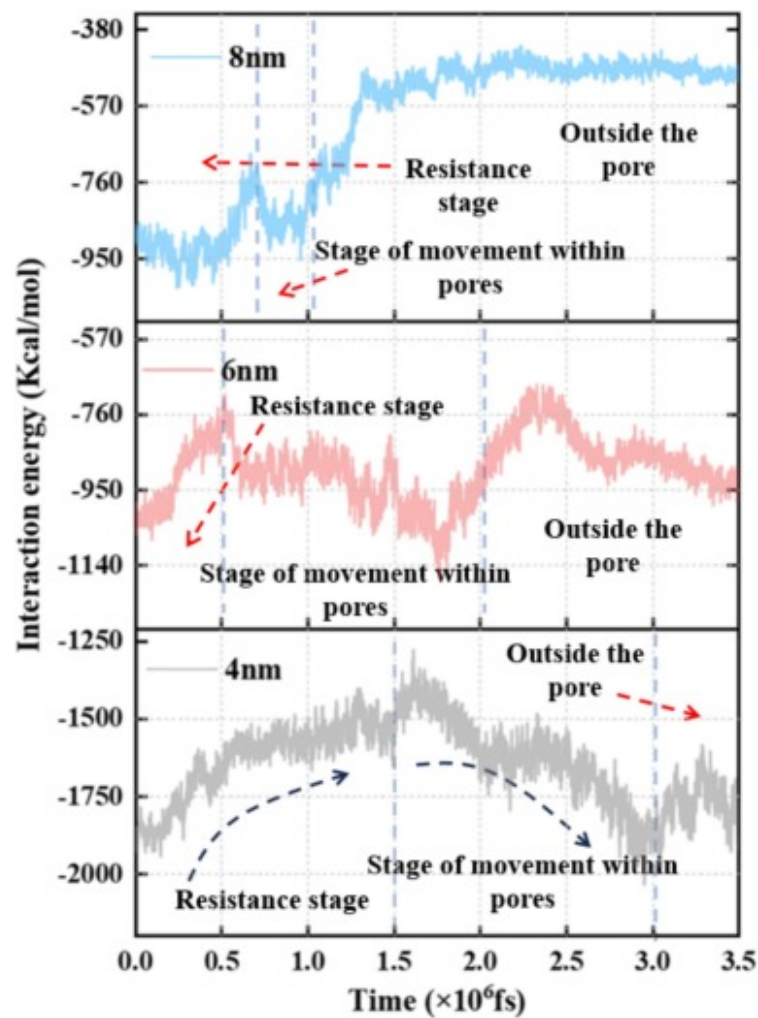
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Fig. 13. Injection characteristics of CO₂ and its competitive wetting with the nanodroplet in the 8nm slit-shaped pore: (a) Centroid variations of graphene and nanodroplet during CO₂ injection; (b) Density distribution of CO₂ within the 8nm slit-shaped pore during CO₂ injection; (c) Gas-water flow characteristics and wettability change mechanism in the 8nm slit-shaped pore.

The changes in coal-water interaction energy during CO₂ injection (calculated by Eq. (5)) explain the migration characteristics of droplets in different-sized pores from an energetic perspective. As presented in Fig. 14, regardless of pore size, an initial decrease

in coal-water interaction intensity occurs during injection, and the decrease is longer and more pronounced in a smaller-sized pore, which reflects that the droplet is attempting to resist CO₂ invasion and maintain its initial adsorption configuration. When the coal-water interaction intensity in different-sized pores decreases to a critical threshold, the droplet is unable to remain adsorbed at its initial location and is compelled to move out of the pore. During its movement, it keeps contacting the front pore wall, which strengthens its interaction with the coal to impede further motion. The extent to which this interaction is strengthened is negatively correlated with pore size. After the main body of the droplet leaves the pore space, only a small cluster of water molecules remains inside the pore, with a plummet of coal-water interaction energy observed in different-sized pores.



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Fig. 14. Coal-water interaction energy in different-sized slit-shaped pores.

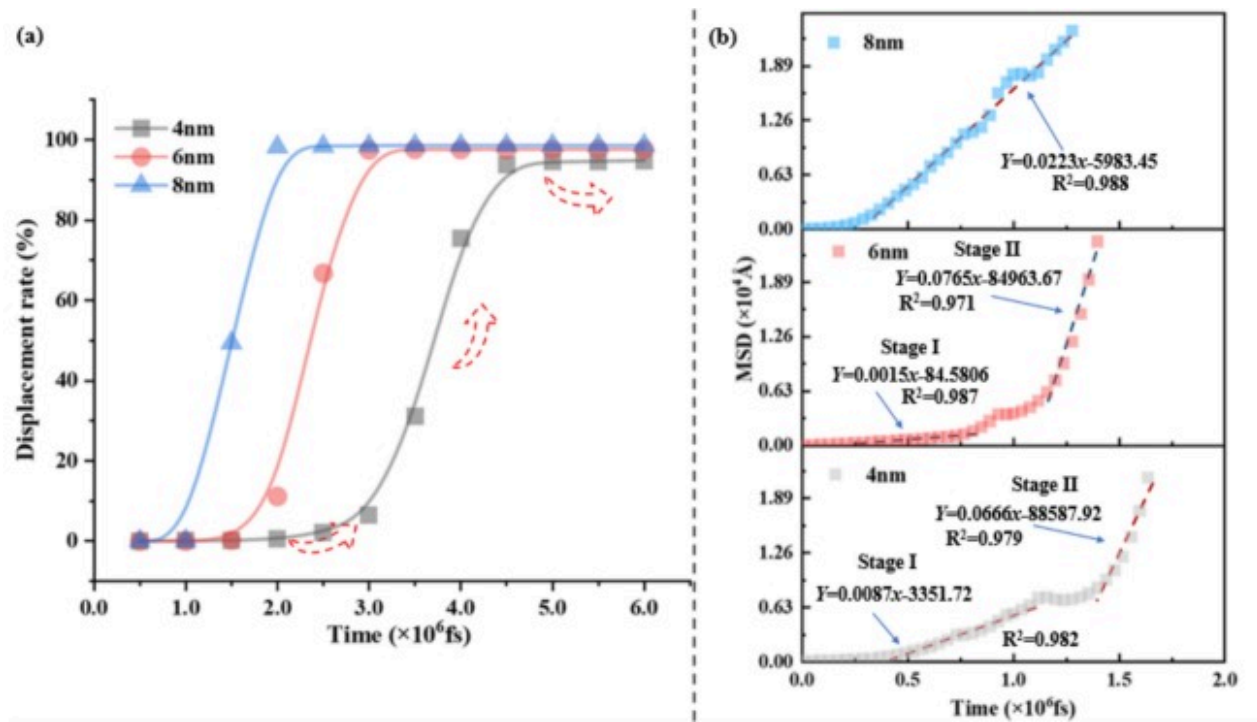
3.3.5. Displacement effect within different-sized slit-shaped pores and activation level of droplets

To reveal the droplet activation level and its relationship with the displacement rate during high-temperature CO₂ injection, the mean square displacements (MSDs) of droplets in different-sized slit-shaped pores were calculated, and point-line graphs

depicting the displacement rates over injection time was plotted (Fig. 15). The MSD and diffusion coefficients were derived by Eq. (6) [57,58]:

$$D = \frac{1}{6N} \lim_{t \rightarrow \infty} \frac{d}{dt} \sum_{i=1} \langle [r_i(t) - r_i(0)]^2 \rangle \quad (6)$$

where N is the number of water molecules; $r_i(t)$ and $r_i(0)$ are atomic position vectors at time t and the initial time, respectively. The droplets adsorbed in pores in a Type I wetting mode have distinct MSDs in the initial resistance stage and the continuous movement stage, the former being obviously smaller than the latter. This finding implies that maintaining continuous droplet movement within pores is far more challenging than detaching droplets from their initial adsorption configurations. A comparison of the MSDs of droplets in small- and medium-sized pores discloses that initiating droplet movement is more difficult in the small-sized pore than in the medium-sized pore, yet continuous droplet movement in the medium-sized pore demands a higher activation level. In contrast, the droplets adsorbed in pores in a Type II wetting mode can detach from their initial adsorption configurations and move continuously at a relatively low activation level.



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Fig. 15. Displacement rates in different-sized slit-shaped pores.

The displacement rates of water molecules in different-sized pores can be derived by [59]:

$$\eta = \left[1 - \frac{N_{H_2O}(t)}{N_{H_2O}(0)} \right] \times 100\% \quad (7)$$

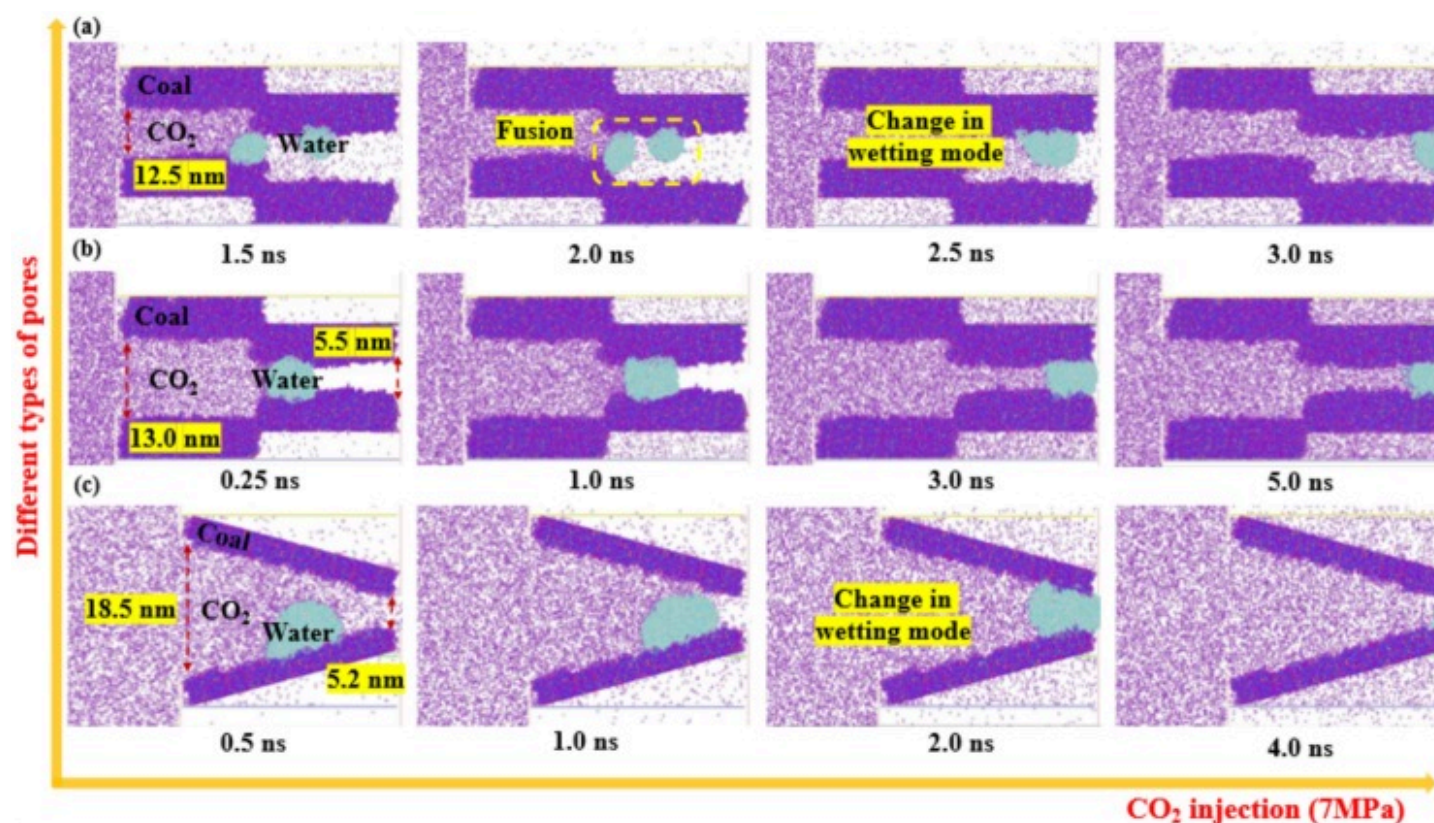
where η is the displacement rate; $N_{H_2O}(0)$ is the initial number of water molecules; $N_{H_2O}(t)$ is the number of water molecules remaining in pores at time t . The displacement rate of water molecules in the large-sized pore is extremely sensitive to high-temperature CO₂. It surges in the initial period of simulation and reaches up to 98.62% upon the completion of injection. The relatively weak water wettability and the special movement mode of droplets in these pores (accompanied by the escape of numerous water molecules) enable them to move out of the pore space after only a slight activation. The droplets in small- and medium-sized pores require a strong activation to leave their initial adsorption sites and to maintain continuous movement. For this reason, the displacement rates of water molecules in small- and medium-sized pores are delayed in response to CO₂ injection, which is more evident in the smaller-sized pore. In addition, upon the completion of CO₂ injection, the displacement rates in pores with diameters of 4.00 nm and 6.00 nm are 94.84% and 97.64% respectively, which confirms greater residual water retention and poorer displacement efficiency in the smaller-sized pore.

3.4. CO₂-water two-phase flow characteristics and dynamic wettability variations in different-shaped connected pores

3.4.1. Migration characteristics of droplets in different-shaped connected pores

Compared to different-sized slit-shaped pores, different-shaped connected pores witness more complicated droplet transport. As displayed in Fig. 16, the wetting behavior of droplets in the Z-shaped pore undergoes phased transformations during high-temperature CO₂ injection. To be specific, the Type II wetting mode transforms to Type I when passing through the pore throat, and it reverts to Type II after the two droplets merge. Owing to the unique existence form of multiple small-volume droplets in the Z-shaped pore, a water lock is created when the droplets pass through the pore throat. Nevertheless, the unstable adsorption configuration is disrupted by a brief impact of high-temperature CO₂. The droplet in the abruptly constricted/expanded pore

maintains the Type I wetting mode throughout the displacement process. The large-volume cavity at the front manages to accommodate the intruding CO₂ molecules, which results in a weakened impact on the droplet in the early period of CO₂ injection and a delayed detachment (evident only at 1.00ns) from the initial adsorption sites. Moreover, the water lock formed by the large-volume droplet in the rear narrow channel is relatively robust and still cannot be completely broken even at 5.00ns. The water lock that forms at the opening of the wedge-shaped pore is weaker than that in the abruptly constricted/expanded pore, and undergoes a more vigorous impact from CO₂. This water lock is almost completely eliminated by 4.00ns.



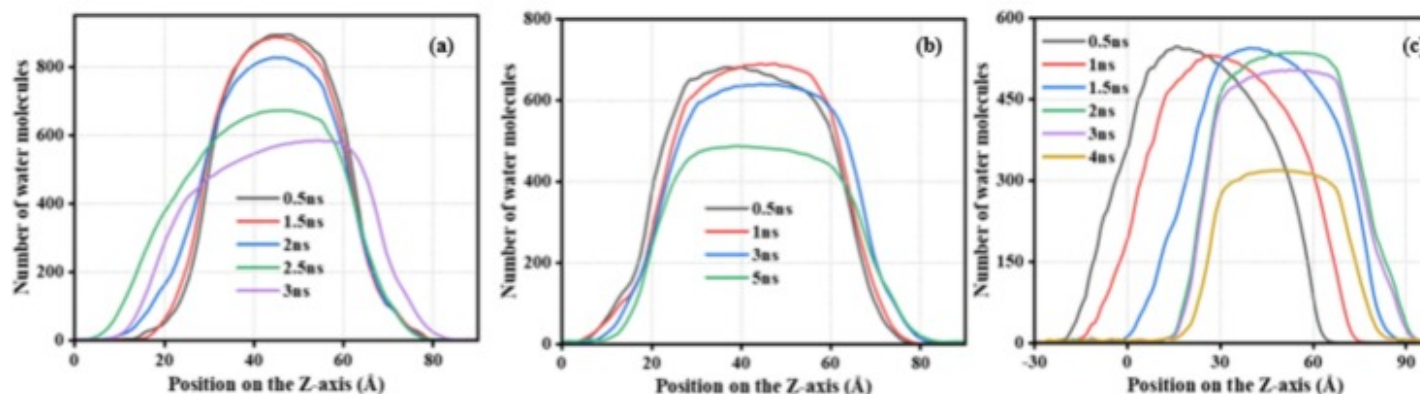
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Fig. 16. Movement of droplets in different-shaped connected pores: a) Z-shaped pore; (b) Abruptly constricted/expanded pore; (c) Wedge-shaped pore.

3.4.2. Morphological changes of droplets in different-shaped connected pores

The spatial distribution of water molecules in different-shaped connected pores in the Z-direction at typical moments is provided in Fig. 17. Obviously, the morphology of droplets in different-shaped connected pores correlates strongly with their wetting modes. In a Type I wetting mode, the number of droplets in the Z-direction is symmetrically distributed, and the peak value appears in the central area of the pores. However, in a Type II wetting mode, the number of droplets in the Z-direction is asymmetrically distributed, and the position where the peak value appears is closely related to the coal-water contact angle. Specifically, the peak value appears near the coal surface when the coal-water contact angle is small, while it appears far from the coal surface when the coal-water contact angle is large. When impacted by CO₂, droplets in a Type II wetting mode are more unstable than those in Type I. They possess a larger advancing contact angle and a smaller receding contact angle during their movement, and such a tilted shape is conducive to the transition from water-wet to gas-wet states. Besides, obvious droplet morphological changes induced by wetting mode alterations are observed in both the Z-shaped pore and the wedge-shaped pore, while the droplet morphological changes in the abruptly constricted/expanded pore primarily stem from the vibration of water molecules in the middle.



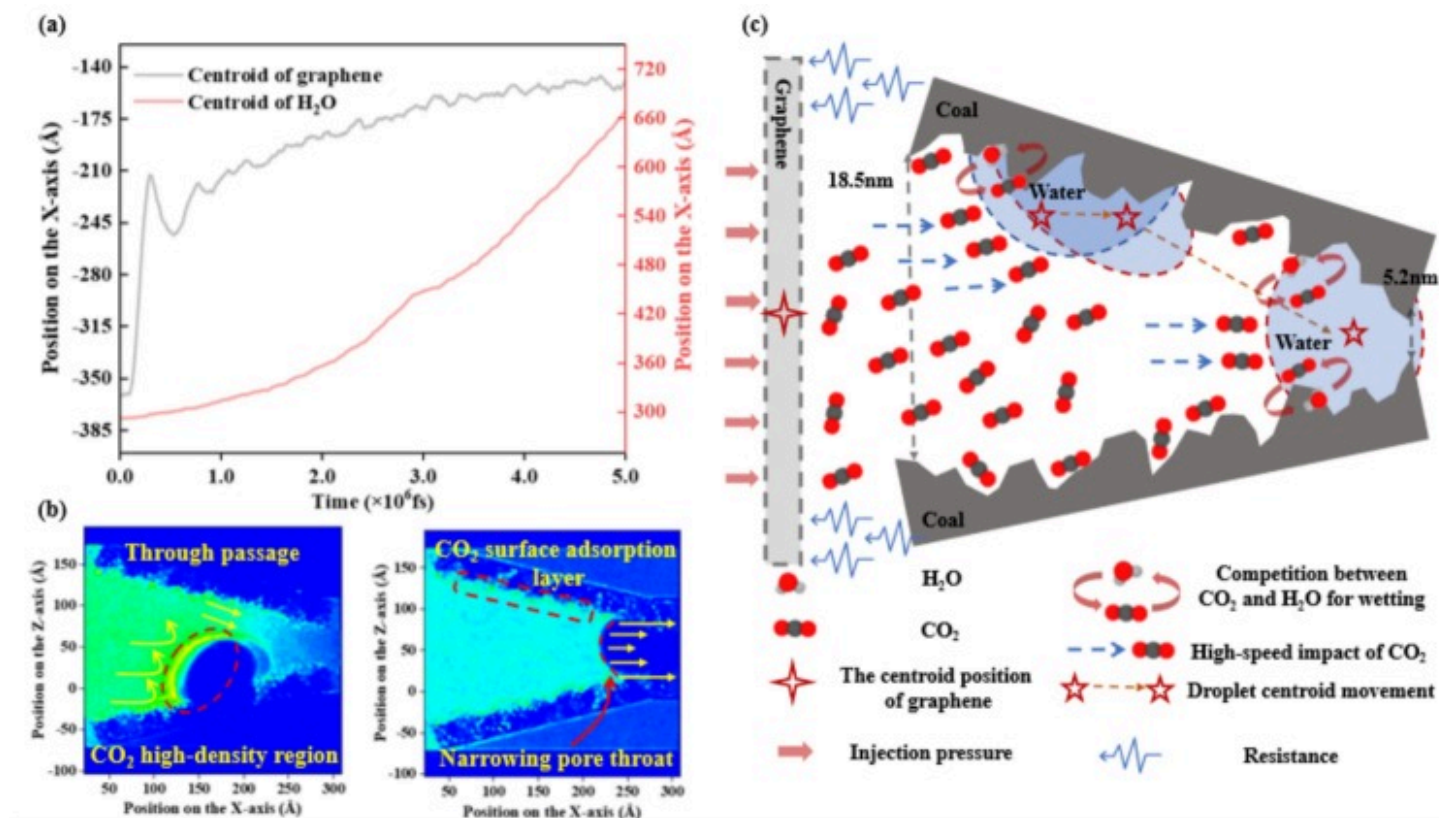
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Fig. 17. Spatial distribution of water molecules in different-shaped connected pores.

3.4.3. Wettability variation and droplet resistance in different-shaped connected pores

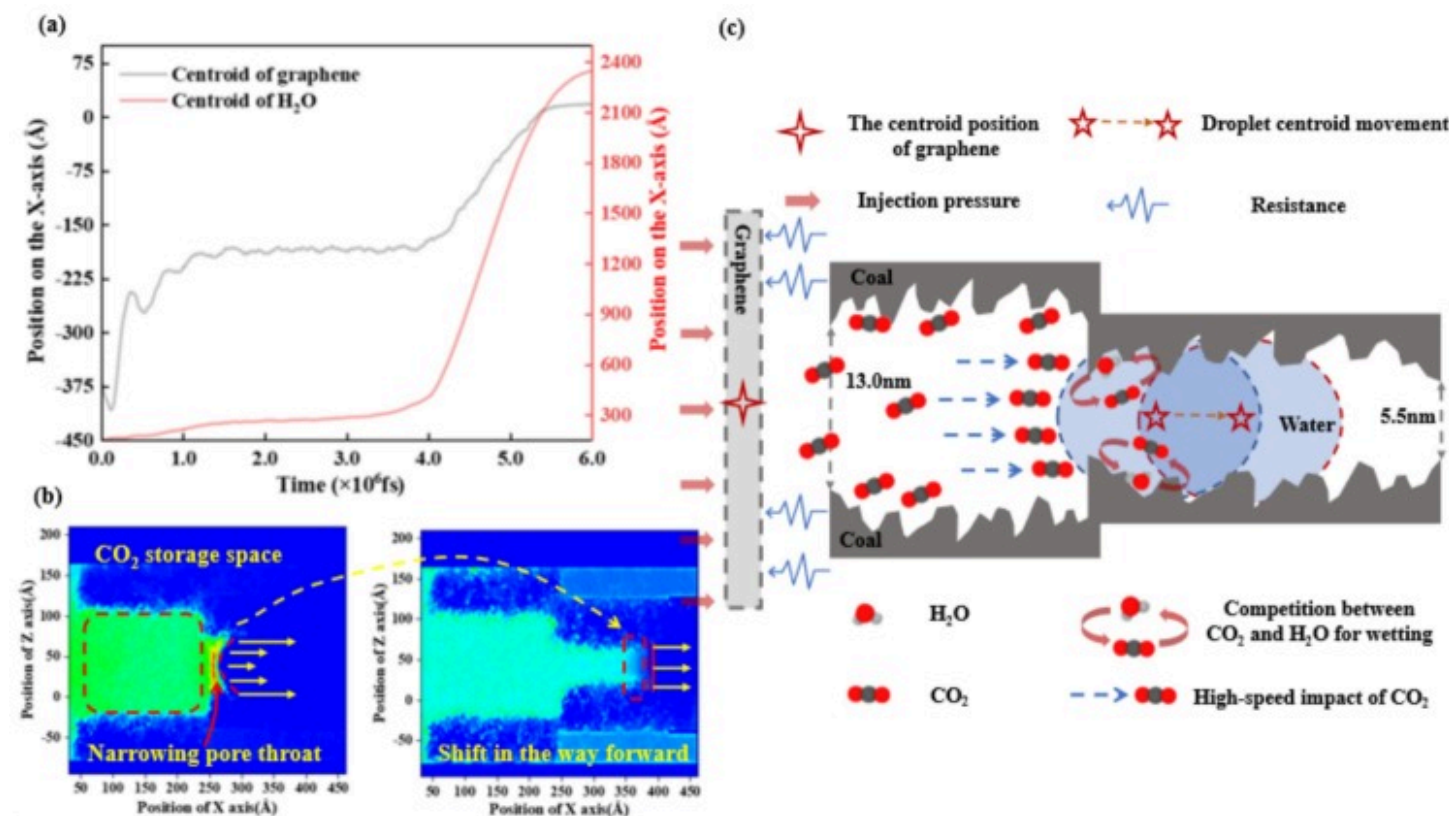
Similar to small- and medium-sized slit-shaped pores, different-shaped connected pores all present piston wall oscillations along the X-direction during the initial period of CO₂ injection, yet with varying intensities. As can be seen from [Fig. 18](#), the injectability of high-temperature and high-pressure CO₂ within the wedge-shaped pore is relatively poor, and the piston wall oscillates violently. This phenomenon is explained as follows: On the one hand, the droplet adsorbed within the wedge-shaped pore in a Type II wetting mode possesses a large volume. On the other hand, the pore diameter decreases continuously in the direction of CO₂ molecule advancement. When passing through such a narrow flow channel at high speeds, substantial CO₂ molecules are forced to rebound upon impacting the droplet, forming a high-pressure zone behind the droplet. The high-pressure zone in turn impedes continuous CO₂ injection. The abruptly constricted/expanded pore and the Z-shaped pore demonstrate better injectability of high-temperature and high-pressure CO₂, as well as milder piston wall oscillations. As given in [Fig. 19](#), [Fig. 20](#), the droplet adsorbed at the throat of the abruptly constricted/expanded pore in a Type I wetting mode or the small-volume droplets adsorbed on the upper and lower walls of the Z-shaped pore in a Type II wetting mode reserves sufficient space for the rapidly advancing CO₂ molecules. In the meantime, the spacious pore cavity constitutes a CO₂ reservoir with a uniform density and a certain pressure-regulating capability, which in turn diminishes the influence of droplet movement on CO₂ molecules outside the pore.



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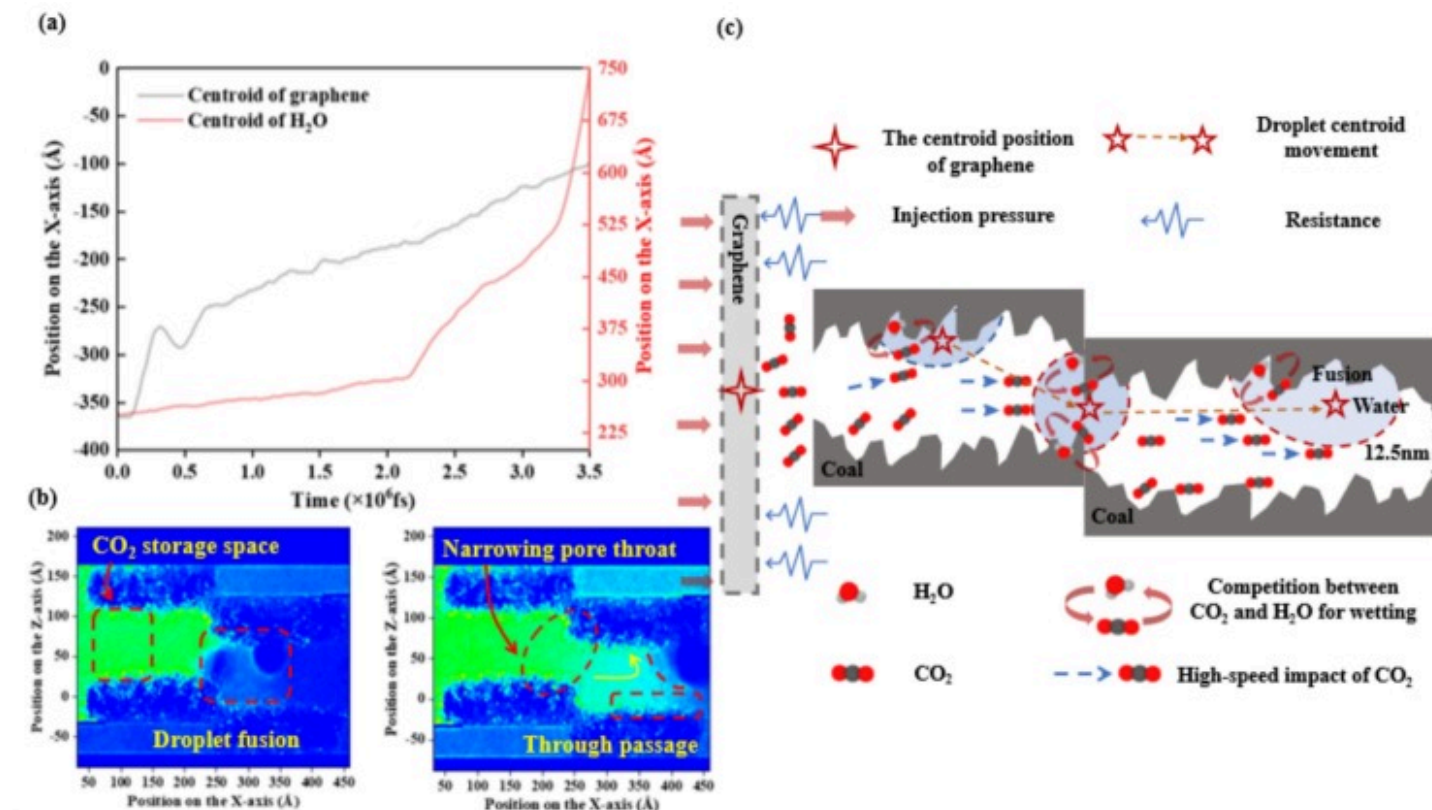
Fig. 18. Injection characteristics of CO₂ and its competitive wetting with the nanodroplet in the slit-shaped pore: (a) Centroid variations of graphene and nanodroplet during CO₂ injection; (b) Density distribution of CO₂ within the slit-shaped pore during CO₂ injection; (c) Gas-water flow characteristics and wettability change mechanism in the slit-shaped pore.



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Fig. 19. Injection characteristics of CO₂ and its competitive wetting with the nanodroplet in the abruptly constricted/expanded pore: (a) Centroid variations of graphene and nanodroplet during CO₂ injection; (b) Density distribution of CO₂ within the abruptly constricted/expanded pore during CO₂ injection; (c) Gas-water flow characteristics and wettability change mechanism in the abruptly constricted/expanded pore.



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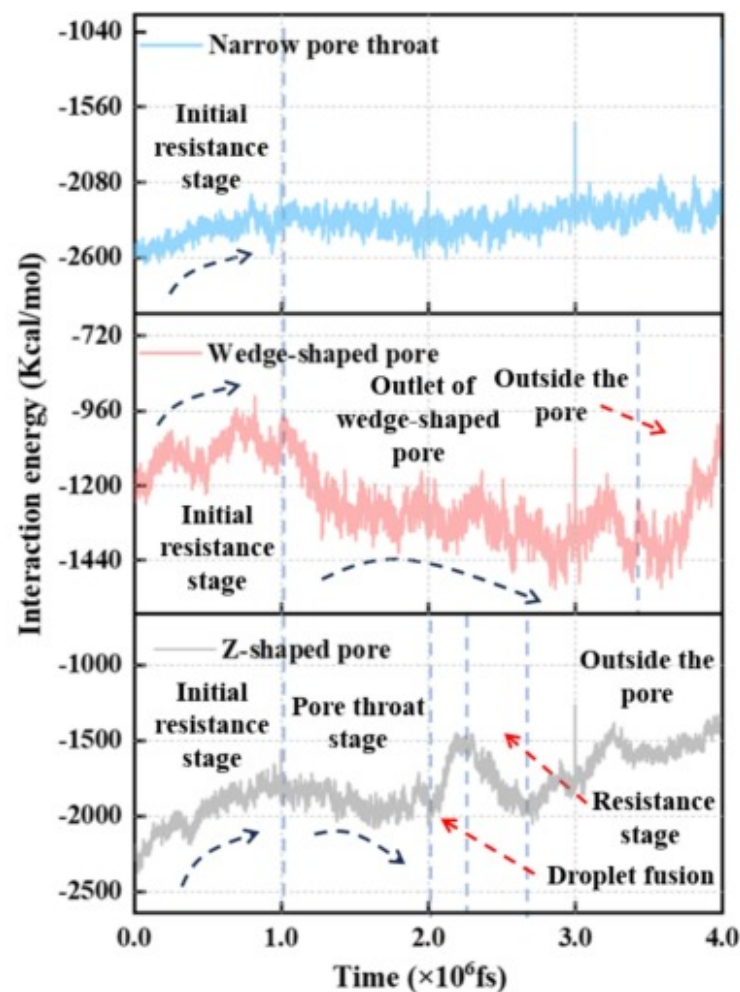
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Fig. 20. Injection characteristics of CO₂ and its competitive wetting with the nanodroplet in the Z-shaped pore: (a) Centroid variations of graphene and nanodroplet during CO₂ injection; (b) Density distribution of CO₂ within the Z-shaped pore during CO₂ injection; (c) Gas-water flow characteristics and wettability change mechanism in the Z-shaped pore.

The wedge-shaped pore maintains a high droplet centroid velocity along the X-direction throughout the injection process, even in the gradual water-lock stage at the pore opening (1.50–4.00 ns). The reason is that water molecules that constitute the water lock at the pore opening keep detaching from the droplet and dispersing into the pore space in this stage, although the strong resistance from the droplet to the transition from water-wet to gas-wet states causes CO₂ to move forward slowly in a crescent shape. In Fig. 19, the abruptly constricted/expanded pore which retains a constant wetting mode and a slightly changed droplet

shape boasts relatively good wettability. The droplet movement there is slow but highly integrated. In [Fig. 20](#), the Z-shaped pore displays distinct transitions of movement speed: The centroid of the droplets moves slowly before the droplets pass through the pore throat in a Type I wetting mode, whereas the movement obviously accelerates after the droplets merge and move on the upper pore wall in a Type II wetting mode.

Among the three types of different-shaped connected pores, the Z-shaped pore witnesses the most complex changes in coal-water interaction intensity. As depicted in [Fig. 21](#), the coal-water interaction intensity shows a cyclic change pattern of weakening-strengthening-weakening-strengthening during continuous CO₂ injection, which is respectively related to droplet resistance to detaching from initial adsorption sites, droplet movement in a Type II wetting mode, and droplet fusion. In the wedge-shaped pore, the gradually forming water lock at the pore opening enables the coal-water interaction intensity to rise continuously for as long as 2.00ns after the droplet starts moving (below the critical threshold). The abruptly constricted/expanded pore maintains a relatively intense coal-water interaction, despite a certain weakening in the initial resistance stage, which explains, from an energy perspective, why the droplet moves slowly under the impact of high-temperature and high-pressure CO₂.



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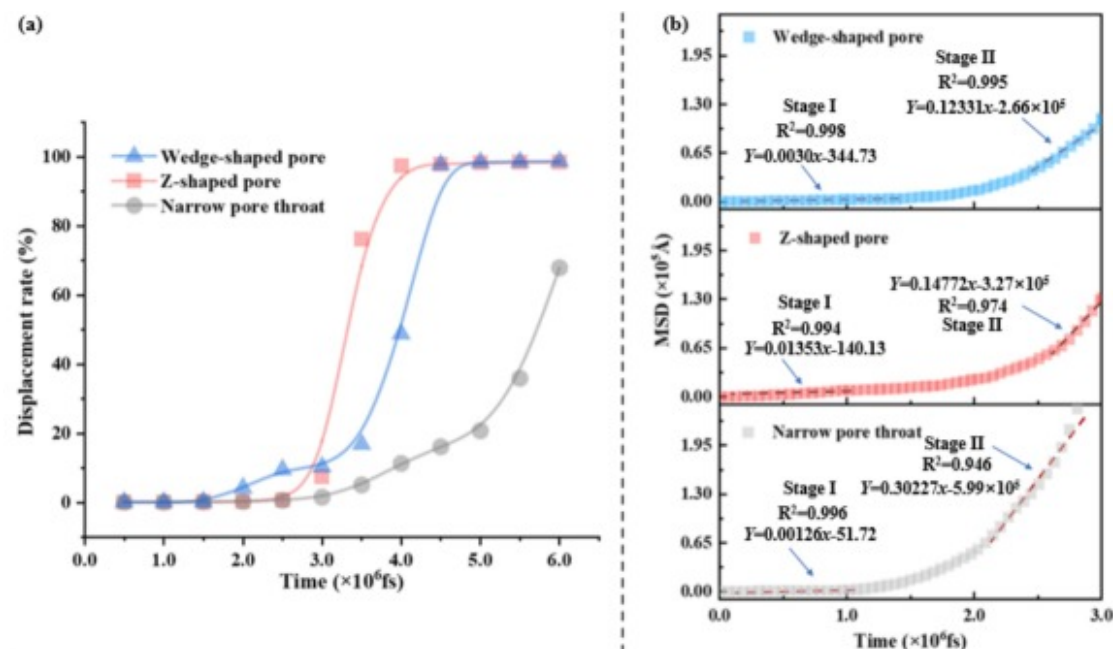
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Fig. 21. Coal-water interaction energy in different-shaped connected pores.

3.4.4. Displacement effect within different-shaped connected pores and activation level of droplets

How the MSDs and displacement rates η of water molecules within different-shaped connected pores (obtained by Eqs. (6), (7), respectively) change over the simulation time is showcased in Fig. 22. In all of the three types of pores, the MSDs of water

molecules can be divided into two stages according to the injection time, corresponding to the droplets' initial resistance and continuous movement in the pores respectively. To better investigate the activation level of droplets, representative areas were selected in the two stages for linear fitting. In the abruptly constricted/expanded pore, the spacious cavity at the front provides a buffer space for the rapidly advancing CO₂ molecules in the initial stage, resulting in a poor activation of the droplet located at the pore throat in the middle and a low displacement rate of water molecules compared to those in other pores in the same period. As CO₂ injection proceeds, the accumulating pressure of CO₂ molecules in the pore cavity at the front continuously provides power for displacing the droplet in the narrow pore cavity at the rear without requiring substantial new CO₂ influx. As a consequence, the droplet in the abruptly constricted/expanded pore receives the strongest activation effect in the second half of the injection, accompanied by a markedly improved displacement rate of water molecules. In the Z-shaped pore, the droplets are more dispersed and of smaller individual volumes. Moreover, the large pore space (excluding the throat) enables the droplets to be impacted by more CO₂ molecules in the initial stage of injection and achieves a better activation effect. However, the displacement rate of water molecules remains limited in this period because of the considerable distance between the activated droplet and the pore opening. After the two droplets merge, the wetting mode remains unchanged, but the displacement rate of water molecules soars. Although the wedge-shaped pore merely receives modest activation from the droplet in the initial resistance stage, the water molecules in it achieves a superior displacement rate in this stage for two reasons: First, the droplet adsorbed on the pore wall in a Type II wetting mode starts to move as a whole under the impact of CO₂. Second, a large number of water molecules on the droplet peripheries take the lead in escaping outside the pore. After the water lock is gradually formed at the pore opening, the altered wetting mode and the intensified activation of the droplet jointly trigger a surge of the displacement rate of water molecules.



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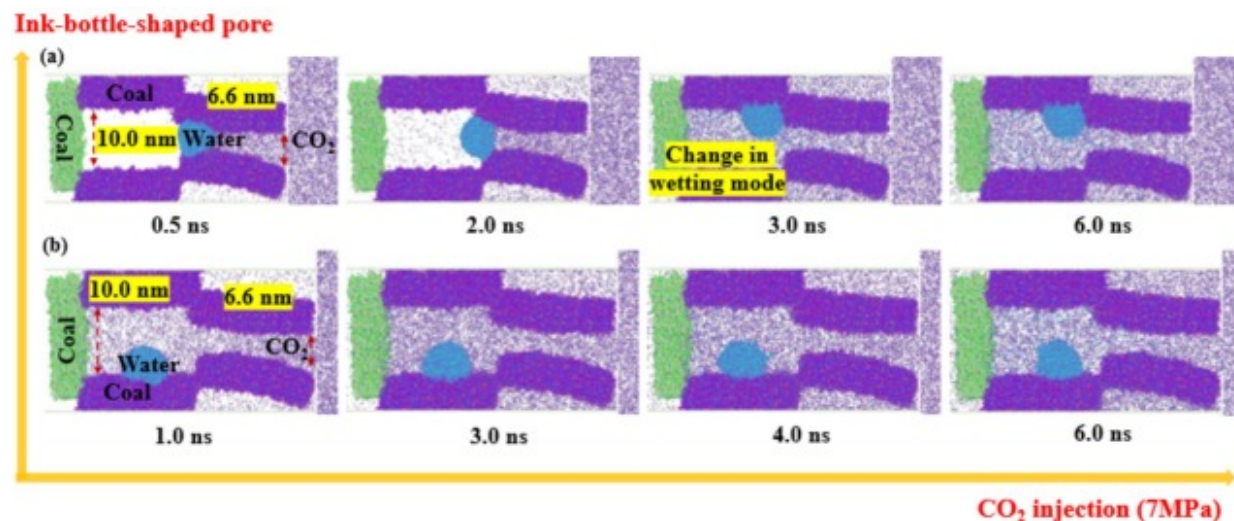
Fig. 22. Displacement rates in different-shaped connected pores.

3.5. CO₂-water two-phase flow characteristics and dynamic wettability variations in semi-closed pores

3.5.1. Migration characteristics of droplets in semi-closed pores

In contrast to connected pores, ink-bottle-shaped pores lack a connected flow channel that permits CO₂ passage, which makes them an ideal space for geological CO₂ sequestration. As injection proceeds, the CO₂ pressure within the pores rises progressively until it equilibrates with the external environment, a process that coincides with wettability variations within the pores. Different modes of water molecule layer placement result in distinct wetting modes of droplets within the pores (Fig. 23). To be specific, the first method, after relaxation, leads to the Type I wetting mode where the water molecule layer forms a water lock at the pore neck, while the second method results in a Type II wetting mode, where the droplet is adsorbed on the rear cavity surface. Under the impact of CO₂, the water lock at the pore neck is disrupted. Accordingly, the droplet moves towards the rear

cavity and ultimately adsorbs at an angle between the upper and side walls of the pore in an intermediate wetting mode between Type I and Type II. The droplet adsorbed on the rear cavity surface oscillates slightly as CO₂ injection progresses. During this process, the coal-water contact angle keeps expanding, and massive water molecules detach from the main body of the droplet and disperse into the cavity. Eventually, the wettability deteriorates.



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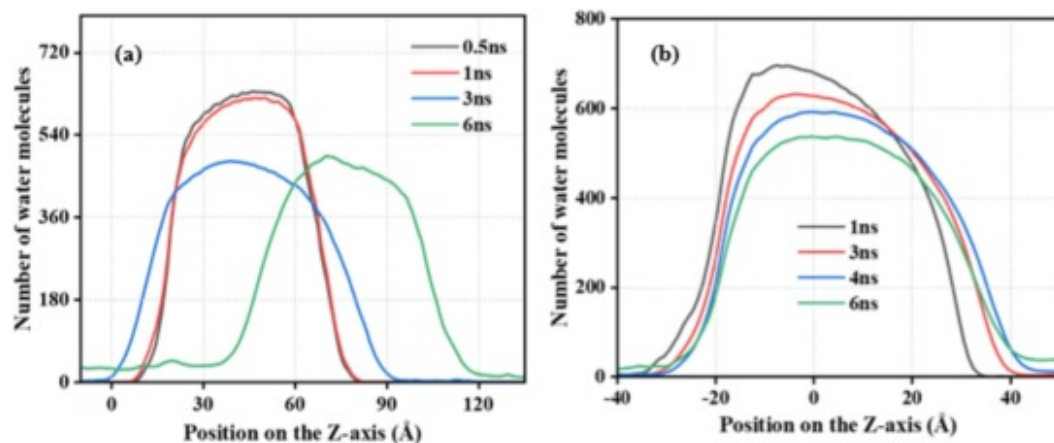
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Fig. 23. Movement of droplets in semi-closed pores: (a) Type I ink-bottle-shaped pore; (b) Type II ink-bottle-shaped pore.

3.5.2. Morphological changes of droplets in semi-closed pores

Fig. 24 presents the distribution of water molecules within the ink-bottle-shaped pore in the Z-direction at typical moments during CO₂ injection under two different modes of water molecule layer placement. The droplet that forms the water lock at the bottleneck experiences a change in its wetting state under the impact of CO₂. In Fig. 24(a), water molecules constituting the droplet maintain a symmetrical distribution after entering the rear cavity, but their distribution becomes more uniform across different heights, indicative of weakened compression received by the droplet. For the droplet adsorbed on the lower pore wall in a Type II wetting mode, CO₂ injection continuously decreases the number of water molecules in the coal-water adsorption layer, triggering a transition into a spherical shape, reducing the coal-water contact area, and deteriorating the wettability. In addition,

both types of droplets undergo noticeable molecular escapes on their surfaces during the middle and late stages of injection (3.00–6.00ns), so that the rear cavity of ink-bottle-shaped pores is filled with water molecules. Driven by high-temperature CO₂, some of these water molecules pass through the bottleneck channel to leave the pore space, thus completing their displacement and the transition from water-wet to gas-wet states. This displacement method differs significantly from the one (where the entire droplet leaves the pore space) in interconnected pores.



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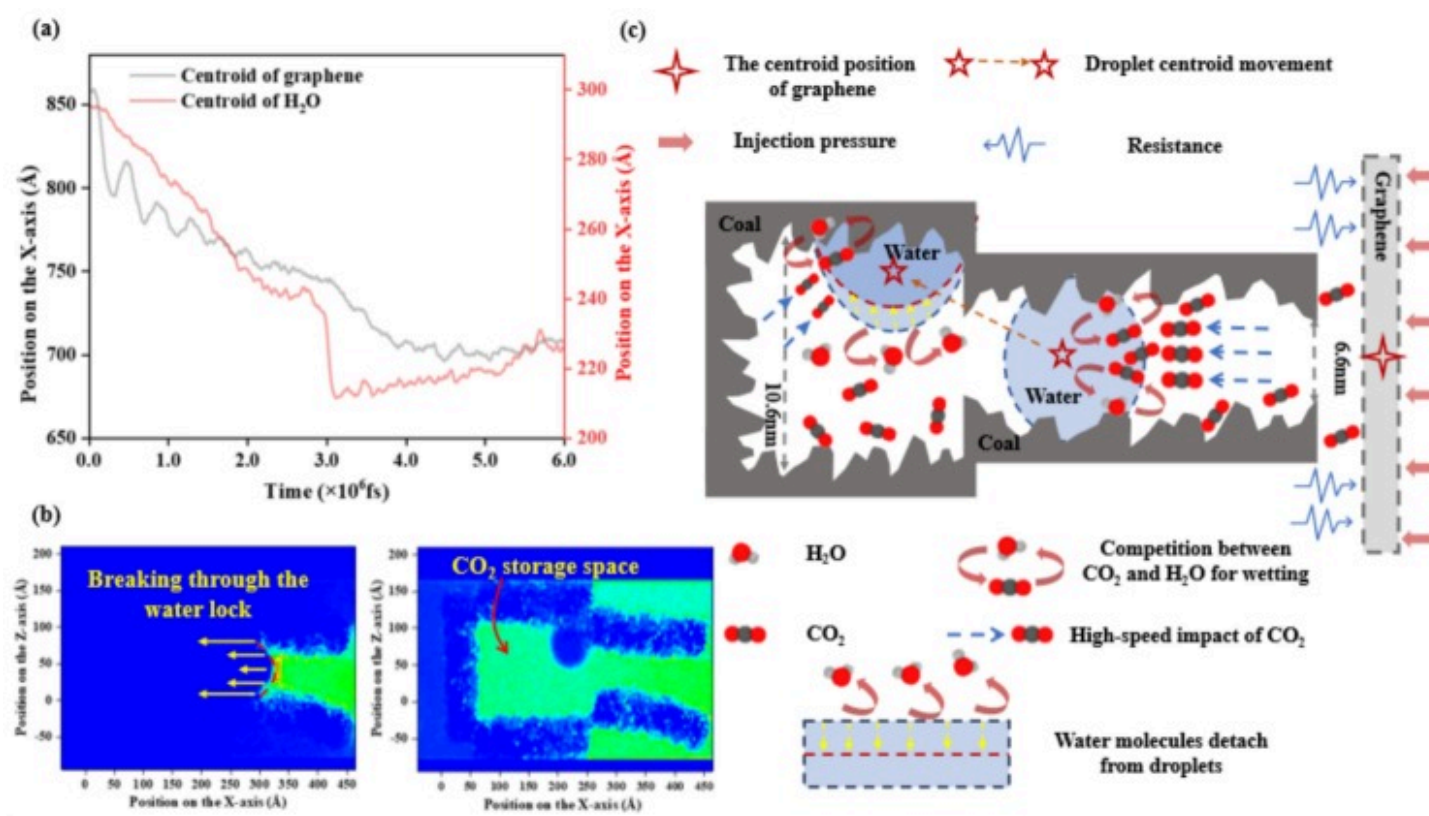
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Fig. 24. Spatial distribution of water molecules in semi-closed pores.

3.5.3. Wettability variation and droplet resistance in semi-closed pores

Similarly, the injectability of CO₂ in ink-bottle-shaped pores is poor in the initial stage (Fig. 25, Fig. 26). Noticeable piston wall oscillations occur in the X-direction, probably because the bottleneck channel is quite narrow. Additionally, the droplet adsorbed at the bottleneck in a Type I wetting mode completely blocks the originally narrow bottleneck channel. As a result, CO₂ must first break the water lock before entering the large-volume rear cavity, which creates additional injection challenges in the initial stage and causes more intense piston wall oscillations in the X-direction. The droplet adsorbed at the bottleneck in the Type I wetting mode is subjected to a stronger impact from CO₂. Affected by the coupling of molecular impact and competitive wetting, high-temperature CO₂ advances in a crescent shape within the narrow bottleneck. The droplet continuously moves leftwards, with its centroid coordinates rapidly decreasing in the X-direction until a sudden change at 3.00ns. At that moment, the water lock is

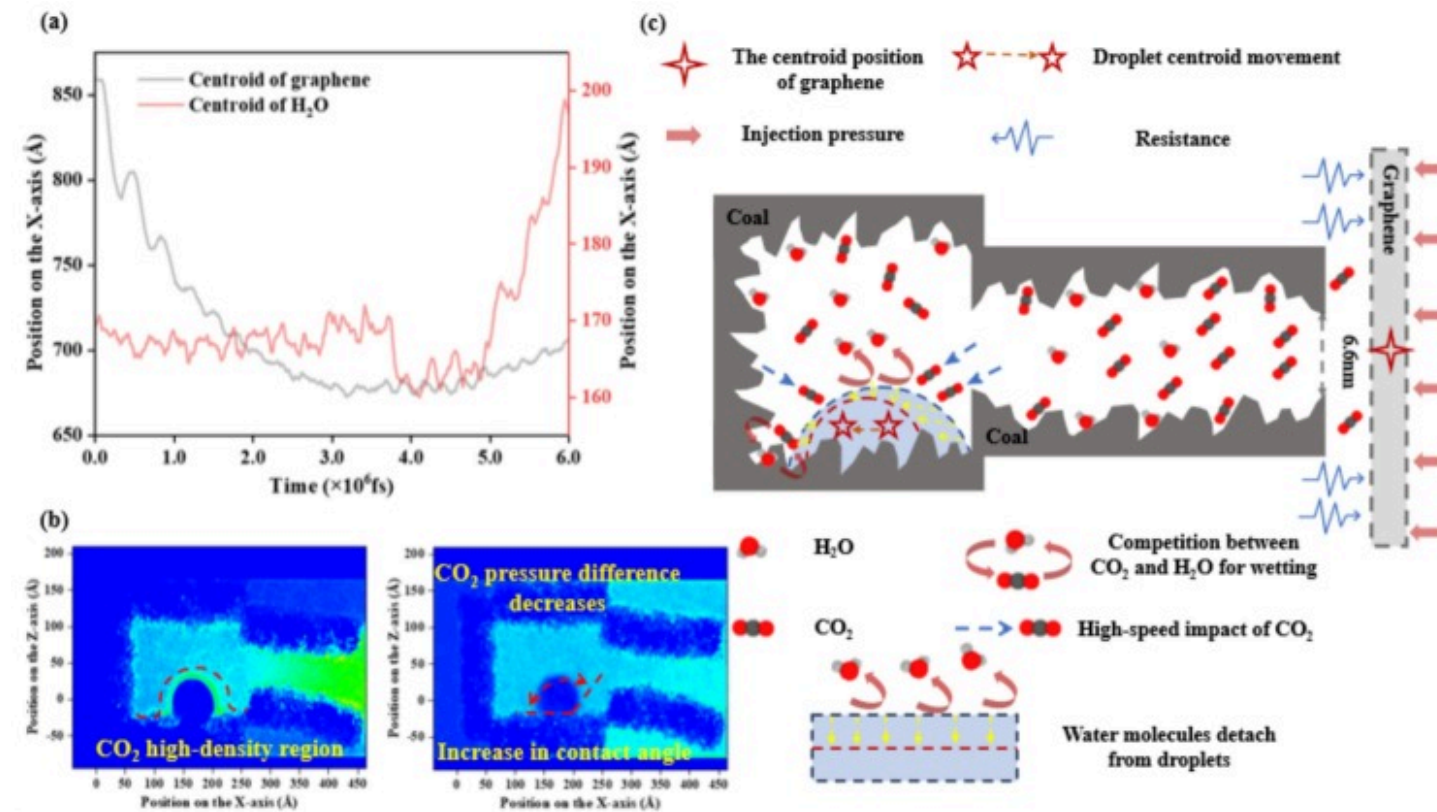
eliminated, and the droplet enters the rear cavity and adsorbs at an angle between the upper and side walls. In the early stage of injection, few CO₂ molecules that manage to enter the rear cavity tend to adsorb onto the pore and droplet surfaces rather than move within the cavity. During this stage, competitive wetting dominates over high-speed impact (Fig. 26), which expands the coal-water contact angle, deteriorates the wettability, and brings about insignificant movement of the droplet (minimal change in centroid coordinates in the X-direction). As injection continues, CO₂ molecules in the cavity become denser, violently impact the droplet adsorbed on the pore surface in a Type II wetting mode, and force it to move overall. The centroid coordinates in the X-direction change suddenly near 3.50ns. In the late stage of injection, massive water molecules escape from the droplet surface into the cavity, which remarkably reduces the droplet volume. Some water molecules, carried by high-temperature CO₂, pass through the bottleneck channel and leave the pore space. Due to such a highly dispersed distribution of water molecules, centroid coordinates in the X-direction experience a sudden increase.



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Fig. 25. Injection characteristics of CO₂ and its competitive wetting with the nanodroplet in the Type I ink-bottle-shaped pore: (a) Centroid variations of graphene and nanodroplet during CO₂ injection; (b) Density distribution of CO₂ within the Type I ink-bottle-shaped pore during CO₂ injection; (c) Gas-water flow characteristics and wettability change mechanism in the Type I ink-bottle-shaped pore.

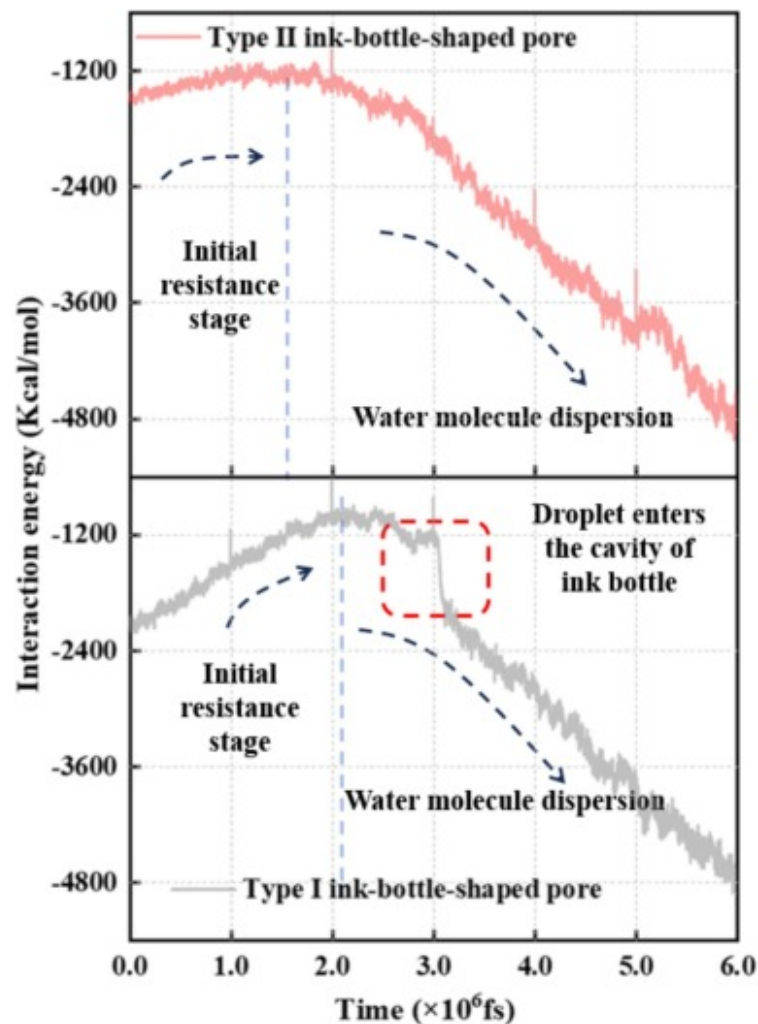


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Fig. 26. Injection characteristics of CO₂ and its competitive wetting with the nanodroplet in the Type II ink-bottle-shaped pore: (a) Centroid variations of graphene and nanodroplet during CO₂ injection; (b) Density distribution of CO₂ within the Type II ink-bottle-shaped pore during CO₂ injection; (c) Gas-water flow characteristics and wettability change mechanism in the Type II ink-bottle-shaped pore.

The variation of coal-water interaction intensity in ink-bottle-shaped pores over injection time differs markedly from that in connected pores. As shown in Fig. 27, the droplet that is adsorbed at the bottleneck in a Type I wetting mode and forms a water lock demonstrates stronger initial resistance to wettability alteration compared to the droplet that is adsorbed in a Type II wetting mode, as it requires a much greater interaction energy reduction to detach from its initial adsorption configuration. However, as CO₂ injection continues, the coal-water interaction intensity no longer decreases but goes up instead. This is due to the fact that only a small proportion of water molecules that disperse from the droplet surface into the cavity are carried out of the pore space by CO₂, while the majority remains adsorbed to the coal surface. This tendency is more prominent in the droplet existing in a Type II wetting mode.



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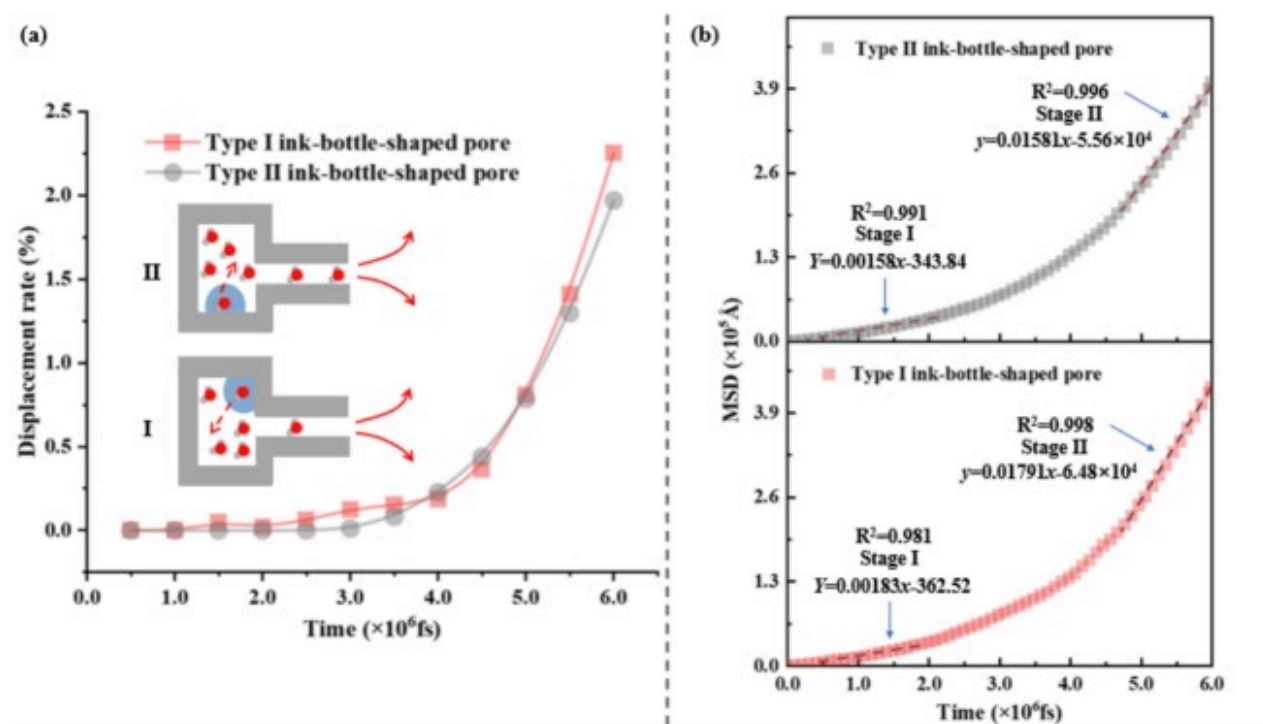
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Fig. 27. Coal-water interaction energy in ink-bottle-shaped pores.

3.5.4. Displacement effect within semi-closed pores and activation level of droplets

How the MSDs and displacement rate η of water molecules within the semi-closed pores (obtained by Eqs. (6), (7), respectively) vary over simulation time is displayed in Fig. 28. Linear fitting on the MSD data unveils that, in both the early and late stages of

injection, the droplet existing in a Type I wetting mode is more strongly activated by high-temperature CO₂ than the one existing in a Type II wetting mode. Correspondingly, its MSD is larger in the same stage. This phenomenon arises because the greater contact area and stronger interaction between the droplet and the pore wall in Type I wetting make it more challenging for the droplet to detach from its initial adsorption configuration and keep moving, which demands an enhanced activation of water molecules constituting the droplet. Besides, the displacement rates of water molecules adsorbed on ink-bottle-shaped pores in both wetting modes follow similar trends over injection time, being merely 2.25% and 1.97% even in the late stages of injection, far lower than the values in connected pores. The reason is that the unique structure of semi-closed pores decides that water molecules can only migrate out of the pore space as dimers or small-volume clusters rather than a large droplet. In this case, a substantial number of water molecules are still trapped within the pore space.



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Fig. 28. Displacement rates in semi-closed pores.

4. Conclusions

In this study, the morphologies and size distribution of coal pores subjected to high-temperature and high-pressure CO₂ treatment were observed with the aid of SEM and AFM. Based on these observations, several typical water-bearing pore models were constructed using molecular dynamics methods. With these models, the gas-liquid flow characteristics and wettability variations during CO₂ injection in different pores were explored. The main conclusions are as follows:

- 1) Droplets exhibit distinct wetting behaviors in pores with different sizes and shapes. Compared to droplets existing a Type II wetting mode, those existing in a Type I wetting mode within the pores possess larger coal-water adsorption layers, stronger coal-water interactions, and better water wettability. Additionally, water wettability is negatively correlated with pore size in a Type II wetting mode, while it is negatively correlated with droplet volume in a Type I wetting mode. For coal reservoirs with small pore sizes and complex pore morphologies, applying a higher CO₂ injection pressure helps to break the strong water-lock effect and improve CBM production. In contrast, for coal reservoirs with wider pore sizes and simpler geometries, using a lower injection pressure may reduce costs and achieve better economic benefits.
- 2) The transition from water-wet to gas-wet states within pores involves two stages, i.e., initial resistance and continuous movement of the droplet, the second stage being more challenging than the first. In the initial period of injection, moderately raising the CO₂ injection pressure conduces to improving the displacement efficiency. The intermolecular hydrogen bonds formed between hydrogen atoms in water and highly electronegative atoms (e.g., nitrogen and oxygen) on pore walls play a crucial role in the wettability of nanodroplets, and pore size critically influences the formation of hydrogen bonds by determining the spatial configuration of nanodroplets within the pores. As the pore size grows, the peak intensity of hydrogen bond distribution plunges, and the distribution pattern transitions from a bimodal pattern to a unimodal pattern. Under high-temperature and high-pressure CO₂ conditions, water molecules in different regions of the droplet are displaced by distinct mechanisms: Competitive wetting dominates for water molecules in the coal-water adsorption layer, whereas high-speed impact prevails in displacing water molecules in the bulk phase of the droplet. In the Type-I wetting mode, the displacement effects of high-speed impact and competitive wetting become more consistent with the increase in pore size. Although competitive wetting

displaces water molecules more effectively than high-speed impact, its lower displacement rate results in a longer time required to eliminate the water-lock effect, particularly in narrower pores.

- 3) The displacement rate of water molecules in pores is related to the mode of transition from water-wet to gas-wet states. In connected pores, droplets move out of the pore space either as a whole or accompanied by the escape of water molecules, leading to fewer residual water molecules in pores and higher displacement rates. In semi-closed pores, only a small portion of water molecules are carried out of the pore space by CO₂ as dimers or small-volume clusters, while most water molecules constituting the droplet remain in the pore space. As a result, their displacement rates are relatively low. In practical injection operations, for coal seams containing numerous non-connected pores, non-connected pores can be transformed into connected ones by moderately raising CO₂ injection pressure and temperature or by employing pulsed injection techniques, so as to boost the efficiency of CBM extraction.

CRedit authorship contribution statement

Shunqing Ma: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology. **Baiquan Lin:** Methodology, Funding acquisition. **Jiajia Zhao:** Software. **Xiangliang Zhang:** Software, Resources, Project administration. **Qian Liu:** Validation, Supervision. **Ting Liu:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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